

Digital Electronics

Previous Year Questions • 1993–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Digital Electronics (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience



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- 3+ years in teaching & the VLSI industry



> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

I

Logic Gates and Boolean Algebra

Topics

1. Introduction to Digital Systems
2. Basic Gates and Concept of Multivibrators Using NOT Gate
3. Universal and Arithmetic Gates
4. Implementing Boolean Expressions Using Universal Logic Gates
5. Timing Diagrams
6. Boolean Algebra, Venn Diagrams, and the Concept of Dual

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Unit Snapshot*

Stream: IN	**Questions:** 25	**MCQ:** 24	**MSQ:** 0	**NAT:** 1
1M: 18	**2M:** 7	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–3M(2023)

**See preface for how Avg Weightage and Range are calculated.*

Q1.25.1 MCQ 2025 1M

A Boolean Function X is given as $X = \bar{A}\bar{B} + \bar{A}\bar{C}$. The reduced form of $\bar{X}$ is:

A. $\bar{A} + \bar{B} + \bar{C}$ **B.** $A + BC$
C. $\bar{A} + \bar{B} + C$ **D.** $B + AC$

Q1.23.1 MCQ 2023 1M

Match the Exclusive-OR (XOR) operations (i) to (iv) with the results (p) to (s) , where X is a Boolean input.

$(i)\ X \oplus X$	$(p)\ 1$
$(ii)\ X \oplus \bar{X}$	$(q)\ 0$
$(iii)\ X \oplus 0$	$(r)\ \bar{X}$
$(iv)\ X \oplus 1$	$(s)\ X$

A. (i) - q , (ii) - r , (iii) - s , (iv) - p

B. (i) - q , (ii) - r , (iii) - p , (iv) - s

C. (i) - p , (ii) - s , (iii) - q , (iv) - r

D. (i) - q , (ii) - p , (iii) - s , (iv) - r

Q1.23.2 MCQ 2023 2M

For the given digital circuit, $A = B = 1$. Assume that AND, OR and NOT gates have propagation delays of 10ns, 10ns and 5ns respectively. All lines have zero propagation delay. Given that $C = 1$ when the circuit is turned on, the frequency of steady state oscillations of the output Y is

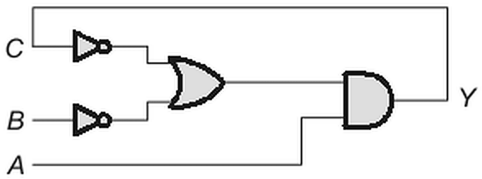


Fig. Q1.23.2

A. 20 MHz **B.** 15 MHz **C.** 40 MHz **D.** 50 MHz

Q1.22.1 MCQ 2022 2M

The logic block shown has an output F given by:

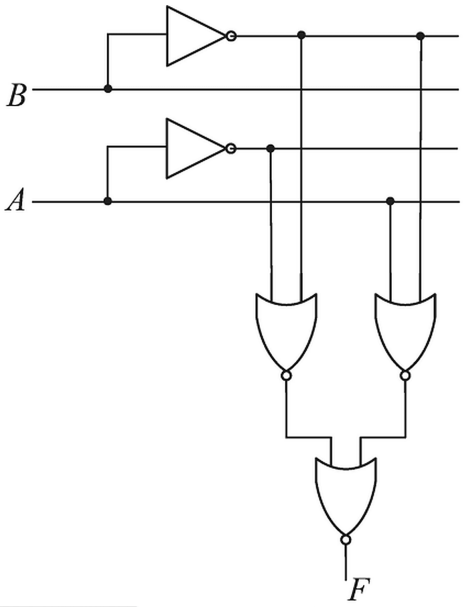


Fig. Q1.22.1

A. $A + B$ **B.** $A.\bar{B}$ **C.** $A + \bar{B}$ **D.** $\bar{B}$

Q1.21.1 MCQ 2021 1M

A Boolean function F of three variables x, y and z is given as:

$F(x, y, z) = (x' + y + z).(x + y' + z').(x' + y + z').(x'y'z' + x'yz' + xyz')$

Which of the following is true?

A. $F(x, y, z) = (x' + y).(x + y' + z')$

B. $F(x, y, z) = (x + y + z).(x' + y' + z')$

C. $F(x, y, z) = x'y'z + xyz$

D. $F(x, y, z) = x'z' + yz'$

Q1.20.1 MCQ 2020 1M

The Boolean expression for the shaded region shown in the figure is:

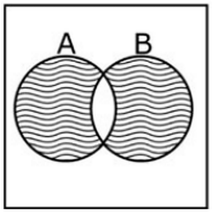


Fig. Q1.20.1

A. $(A + \bar{B}).(\bar{A} + B)$ **B.** $(\bar{A} + \bar{B}).(A + \bar{B})$
C. $(\bar{A} + B).(\bar{A} + \bar{B})$ **D.** $(A + B).(\bar{A} + \bar{B})$

Q1.19.1 NAT 2019 1M

The total number of Boolean functions with distinct truth tables that can be defined over 3 Boolean variables is ____ .

Q1.18.1 MCQ 2018 2M

The Boolean function $F(X, Y)$ realized by the given circuit is

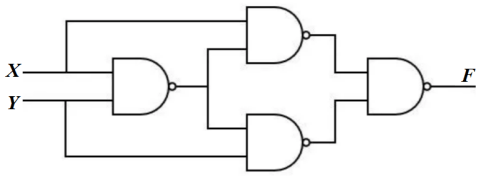


Fig. Q1.18.1

A. $\bar{X}Y + X\bar{Y}$ **B.** $\bar{X}\bar{Y} + XY$ **C.** $X + Y$ **D.** $\bar{X}.\bar{Y}$

Q1.17.1 MCQ 2017 2M

A and B are the logical inputs and X is the logical output shown in the figure. The output X is related to A and B by:

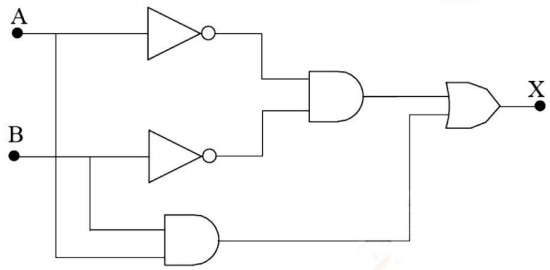


Fig. Q1.17.1

A. $X = \bar{A}B + \bar{B}A$ **B.** $X = AB + \bar{B}A$
C. $X = AB + \bar{B}\bar{A}$ **D.** $X = \bar{A}\bar{B} + \bar{B}A$

Q1.16.1 MCQ 2016 1M

The Boolean expression $XY + (X' + Y')Z$ is equivalent to

A. $XYZ' + X'Y'Z$ **B.** $X'Y'Z + XYZ$
C. $(X + Z)(Y + Z)$ **D.** $(X' + Z)(Y' + Z)$

Q1.15.1 MCQ 2015 1M

Consider the logic circuit with input signal TEST shown in the figure. All gates in the figure shown have identical non-zero delay. The signal TEST which was at logic LOW is switched to logic HIGH. The output

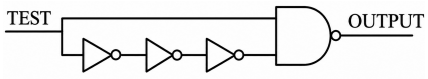


Fig. Q1.15.1

A. Stays HIGH throughout

B. Stays LOW throughout

C. Pulses from LOW to HIGH to LOW

D. Pulses from HIGH to LOW to HIGH

Q1.15.2 MCQ 2015 2M

The logic evaluated by the circuit at the output is:

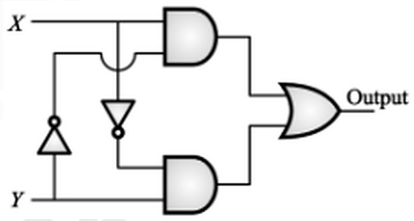


Fig. Q1.15.2

A. $X\bar{Y} + Y\bar{X}$ **B.** $\overline{(X + Y)}XY$
C. $\overline{(XY)}XY$ **D.** $X\bar{Y} + \bar{X}Y + X + Y$

Q1.13.1 MCQ 2013 1M

A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles

A. an AND gate **B.** an OR gate
C. an XOR gate **D.** a NAND gate

Q1.10.1 MCQ 2010 2M

The logic gate circuit shown in the figure realizes the function

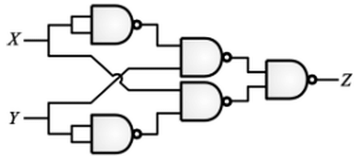


Fig. Q1.10.1

A. XOR **B.** XNOR **C.** Half Adder **D.** Full Adder

Q1.07.1 MCQ 2007 2M

Two square waves of equal period T , but with a time delay τ are applied to a digital circuit, as shown below:

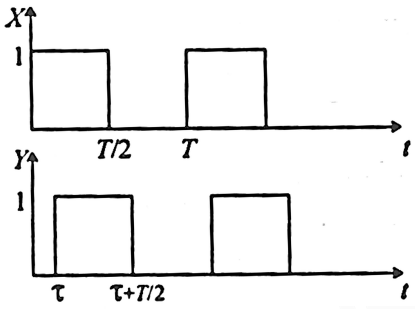
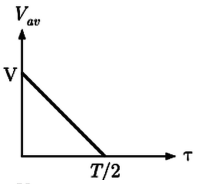


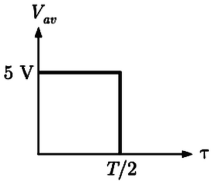
Fig. Q1.07.1

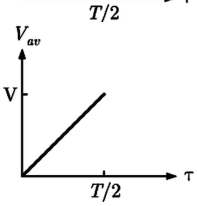
Truth table of the digital circuit is:

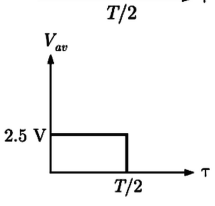
X	Y	Output
0	0	1
0	1	0
1	0	0
1	1	1

The high and the low levels of the output of the digital circuit are 5V and 0V, respectively. Which one of the following figure shows the correct variation of the average value of the output voltage with τ for $0 < \tau < \frac{T}{2}$?

A.


B.


C.


D.


Q1.05.1 MCQ 2005 1M

The expression $Y = \overline{AB}$ is equivalent to

A. $\bar{A} + \bar{B}$ **B.** $AB + A$ **C.** $A + B$ **D.** AB

Q1.05.2 MCQ 2005 1M

Notwithstanding overflow, the addition and subtraction of K-bit signed binary number can be realized using One K-bit full adder and

A. K 2-Input AND Gates

B. K 2-Input NOR Gates

C. K 2-Input OR Gates

D. K 2-Input XOR Gates

Q1.02.1 MCQ 2002 1M

The inverters in the ring oscillator circuit shown below are identical. If the output waveform has a frequency of 10 MHz, the propagation delay of each inverter is:

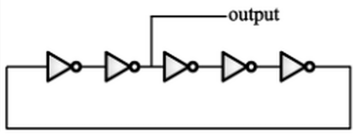


Fig. Q1.02.1

A. 5 ns **B.** 10 ns **C.** 20 ns **D.** 50 ns

Q1.01.1 MCQ 2001 1M

In the logic circuit shown in figure, the output X is

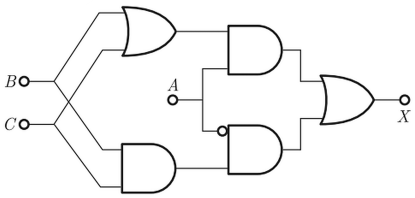


Fig. Q1.01.1

A. $A\bar{B} + B\bar{C} + C\bar{A}$ **B.** $A + B + C$

C. $AB + BC + CA$ **D.** $\overline{AB} + \overline{BC} + \overline{CA}$

Q1.00.1 MCQ 2000 1M

The expression $A + \bar{A}B$ is equivalent to:

A. $A + B$ **B.** $AB + A$ **C.** $A + \bar{B}$ **D.** AB

Q1.98.1 MCQ 1998 1M

For the logic circuit shown in figure, write the expression for Z :

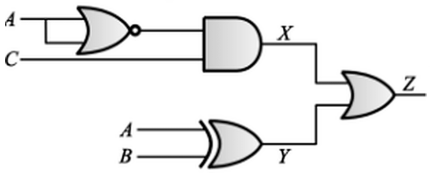


Fig. Q1.98.1

A. $\bar{A}C + A\bar{B} + \bar{A}B$ **B.** $\bar{A}C + A\bar{B}$
C. $AC + AB$ **D.** $AC + BC + AB$

Q1.95.1 MCQ 1995 1M

If A and B are inputs to a logic gate, then match the logic with its output

(a) NAND	(i) $\overline{\bar{A} + \bar{B}}$
(b) NOR	(ii) $\bar{A} + \bar{B}$
(c) XNOR	(iii) $\bar{A}\bar{B} + AB$
(d) AND	(iv) $\bar{A}.\bar{B}$

A. a - ii , b - i , c - iii , d - iv

B. a - ii , b - iv , c - iii , d - i

C. a - i , b - iv , c - ii , d - iii

D. a - iii , b - ii , c - iv , d - i

Q1.94.1 MCQ 1994 1M

The Boolean expression $\overline{A + B + C}$ is equal to

A. $\bar{A} + \bar{B} + \bar{C}$ **B.** $\bar{A}.\bar{B}.\bar{C}$
C. $A + B + C$ **D.** $A.(B + C)$

Q1.94.2 MCQ 1994 1M

Any Boolean function can be realised using only

A. NAND gate **B.** AND gate **C.** OR gate **D.** NOT gate

Q1.92.1 MCQ 1992 1M

In the digital circuit shown below, the output f is found to be logic 1 when A is logic 0. The values of B and C are:

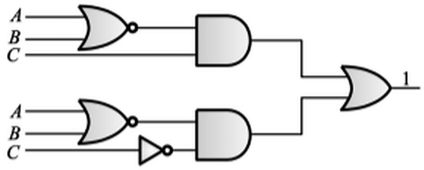


Fig. Q1.92.1

A. $B = 1, C = 0$ **B.** $B = 1, C = 0$ or 1
C. $B = 1, C = 1$ **D.** Indeterminate

Answer Key

Note: **MTA** Marks To All awarded — see detailed solutions.

Q1.25.1 B	Q1.23.1 D	Q1.23.2 A
Q1.22.1 D	Q1.21.1 D	Q1.20.1 D
Q1.19.1 256	Q1.18.1 A	Q1.17.1 C
Q1.16.1 C	Q1.15.1 D	Q1.15.2 A
Q1.13.1 C	Q1.10.1 A	Q1.07.1 A
Q1.05.1 A	Q1.05.2 D	Q1.02.1 B
Q1.01.1 C	Q1.00.1 A	Q1.98.1 A
Q1.95.1 B	Q1.94.1 B	Q1.94.2 A
Q1.92.1 MTA		

UNIT

II

Representation of Boolean Expressions and K-Maps

Topics

1. Minterms and Maxterms
2. SOP and POS Representation
3. Karnaugh Maps (K-Maps)
4. Concept of Implicants

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Unit Snapshot*

Stream: IN	**Questions:** 18	**MCQ:** 17	**MSQ:** 0	**NAT:** 1
1M: 18	**2M:** 0	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2024)–1M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q2.24.1 MCQ 2024 1M

Among the given options, the simplified form of the Boolean function $F = (A + \bar{A}.B) + \bar{A}.(A + \bar{B}).C$ is

A. $A + B + C$ **B.** $A.B.C$
C. $B + \bar{A}.C$ **D.** $\bar{A} + B.C$

Q2.23.1 MCQ 2023 1M

The simplified form of the Boolean function $F(W, X, Y, Z)$, given by

$$F = \Sigma m(4, 5, 10, 11, 12, 13, 14, 15),$$

with the minimum number of terms and smallest number of literals in each term is:

A. $WX + \bar{W}X\bar{Y} + W\bar{X}Y$ **B.** $WX + WY + X\bar{Y}$
C. $X\bar{Y} + WY$ **D.** $\bar{X}Y + \bar{W}\bar{Y}$

Q2.22.1 MCQ 2022 1M

$A = a_1a_0$ and $B = b_1b_0$ are two 2-bit unsigned binary numbers. If $F(a_1, a_0, b_1, b_0)$ is a Boolean function such that $F = 1$ only when $A > B$, and $F = 0$ otherwise, then F can be minimized to the form

A. $a_1\bar{b_1} + a_1a_0\bar{b_0} + a_0\bar{b_0}b_1$

B. $a_1\bar{b_1} + a_1a_0\bar{b_0} + a_0\bar{b_0}\bar{b_1}$

C. $a_1\bar{b_1} + a_1a_0\bar{b_0}$

D. $a_1a_0\bar{b_0} + a_0\bar{b_0}\bar{b_1}$

Q2.20.1 MCQ 2020 1M

Given $f(A, B, C, D) = \Sigma m(0, 1, 2, 6, 8, 9, 10, 11) + \Sigma d(3, 7, 14, 15)$ is a Boolean function, where m represents minterms and d represents don't-cares. The minimal SOP expression for f is:

A. $\bar{D} + A$ **B.** $\bar{A}B + \bar{C}D$
C. $\bar{B} + C$ **D.** $\bar{A}B + CB$

Q2.18.1 MCQ 2018 1M

The product of sum expression of a Boolean function $F(A, B, C)$ of 3 variables is given by

$$F = (A + B + \bar{C}).(A + \bar{B} + \bar{C}).(\bar{A} + B + C).(\bar{A} + \bar{B} + \bar{C})$$

The canonical SOP expression of $F(A, B, C)$ is:

A. $\bar{A}\bar{B}C + \bar{A}BC + A\bar{B}\bar{C} + ABC$

B. $\bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C + AB\bar{C}$

C. $AB\bar{C} + A\bar{B}\bar{C} + \bar{A}BC + \bar{A}\bar{B}\bar{C}$

D. $\bar{A}\bar{B}\bar{C} + \bar{A}BC + AB\bar{C} + ABC$

Q2.12.1 MCQ 2012 1M

In the SOP function $f(X, Y, Z) = \Sigma m(2, 3, 4, 5)$. The prime implicants are:

A. $\bar{X}Y, X\bar{Y}$ **B.** $\bar{X}Y, X\bar{Y}\bar{Z}, X\bar{Y}Z$
C. $\bar{X}Y\bar{Z}, \bar{X}YZ, X\bar{Y}$ **D.** $\bar{X}Y\bar{Z}, \bar{X}YZ, X\bar{Y}\bar{Z}, X\bar{Y}Z$

Q2.11.1 MCQ 2011 1M

For the Boolean expression $f = \bar{a}\bar{b}\bar{c} + \bar{a}b\bar{c} + a\bar{b}\bar{c} + abc + ab\bar{c}$, the minimized POS expression is

A. $f = (b + \bar{c})(a + \bar{c})$ **B.** $f = (\bar{b} + c)(\bar{a} + c)$
C. $f = (\bar{b} + c)(a + \bar{c})$ **D.** $f = \bar{c} + abc$

Q2.09.1 MCQ 2009 1M

The minimal SOP expression for the logic function f represented by the given K-Map is:

RS \ PQ	00	01	11	10
00	0 (0)	1 (1)	0 (3)	0 (2)
01	0 (4)	1 (5)	1 (7)	1 (6)
11	1 (12)	1 (13)	1 (15)	0 (14)
10	0 (8)	0 (9)	1 (11)	0 (10)

Fig. Q2.09.1

A. $QS + P\bar{R}S + PQR + \bar{P}RS + \bar{P}Q\bar{R}$

B. $Q\bar{S} + \bar{P}R\bar{S} + \bar{P}\bar{Q}R + P\bar{R}\bar{S} + P\bar{Q}R$

C. $\bar{P}R\bar{S} + \bar{P}\bar{Q}\bar{R} + P\bar{R}\bar{S} + P\bar{Q}R$

D. $P\bar{R}S + PQR + \bar{P}RS + \bar{P}Q\bar{R}$

Q2.08.1 MCQ 2008 1M

The minimum SOP form of the Boolean expression $y = \bar{P}\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}\bar{S} + P\bar{Q}\bar{R}S + P\bar{Q}RS + P\bar{Q}R\bar{S} + \bar{P}\bar{Q}R\bar{S}$ is:

A. $y = P\bar{Q} + \bar{Q}\bar{S}$ **B.** $y = P\bar{Q} + \bar{Q}R\bar{S}$
C. $y = P\bar{Q} + \bar{Q}\bar{R}\bar{S}$ **D.** $y = \bar{Q}\bar{S} + P\bar{Q}R$

Q2.07.1 MCQ 2007 1M

A logic circuit implements the Boolean function $f = \bar{x}.y + x.\bar{y}.\bar{z}$. It is found that the input combination $x = y = 1$ can never occur. Taking this into account, a simplified expression for f is given by

A. $\bar{x} + \bar{y}.\bar{z}$ **B.** $x + z$
C. $x + y$ **D.** $y + x.\bar{z}$

Q2.06.1 MCQ 2006 1M

Minterm(SOP) expression for a Boolean function is given as:

$$f(A, B, C) = \Sigma m(0, 1, 2, 3, 5, 6)$$

where A is the MSB and C is the LSB. The minimized expression for the function is:

A. $A + (B \oplus C)$ **B.** $(A \oplus B) + C$
C. $\bar{A} + (B \oplus C)$ **D.** $\overline{ABC}$

Q2.05.1 MCQ 2005 1M

A combinational logic circuit has three inputs A, B, C an one output Y . The output $Y = 1$ when at least 2 inputs are 1, otherwise $Y = 0$. In its minimzed SOP realization, the maximum number of 2-input terms is:

A. 1 **B.** 2 **C.** 3 **D.** 4

Q2.04.1 MCQ 2004 1M

The simplest form of the Boolean expression $AB\bar{C}\bar{D} + AB\bar{C}D + ABC\bar{D} + ABCD$ is

A. AD **B.** BC **C.** $\bar{A}B$ **D.** AB

Q2.03.1 MCQ 2003 1M

The Karnaugh Map for a 4-variable Boolean function is given below. The correct Boolean SOP is:

RS \ PQ	00	01	11	10
00	0 (0)	0 (1)	0 (3)	0 (2)
01	1 (4)	0 (5)	0 (7)	1 (6)
11	1 (12)	0 (13)	0 (15)	1 (14)
10	0 (8)	1 (9)	0 (11)	0 (10)

Fig. Q2.03.1

A. $PQRS + \bar{Q}S$ **B.** $\bar{P}QR\bar{S} + \bar{Q}S$
C. $PQR + Q\bar{S}$ **D.** $PQRS + \bar{Q}$

Q2.00.1 MCQ 2000 1M

From the table, choose the correct logical expression for Q:

A	0	0	0	0	1	1	1	1
B	0	0	1	1	0	0	1	1
C	0	1	0	1	0	1	0	1
Q	0	0	0	1	0	1	1	1

A. $AB + BC + CA$ **B.** $A + B + C$
C. $A\bar{B} + B\bar{C} + C\bar{A}$ **D.** $\overline{AB} + \overline{BC} + \overline{CA}$

Q2.98.1 MCQ 1998 1M

The minimal Sum of Products form of $f = A\bar{B}CD + \bar{A}BC + BCD + \bar{A}\bar{B}C$ is

A. $\bar{A}C + BD$ **B.** $\bar{A}C + CD$
C. $AC + \bar{B}D$ **D.** $A\bar{B} + C\bar{D}$

Q2.97.1 NAT 1997 1M

The minimum number of gates used to design the logic system shown in figure satisfying the given truth table is ——

A	B	Y
0	0	$\bar{X}$
0	1	X
1	0	1
1	1	0

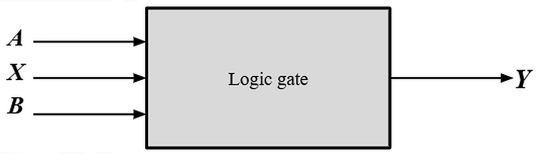


Fig. Q2.97.1

Q2.95.1 MCQ 1995 1M

A combination circuit has inputs A, B and C and its Karnaugh Map is given in figure.

C′ \ A′B′	A′B	AB	AB′
	1 (1)	(3)	1 (2)
C	1 (4)	(5)	1 (7)	(6)

Fig. Q2.95.1

The output of the circuit is:

A. $(A\bar{B} + \bar{A}B)C$ **B.** $(A\bar{B} + \bar{A}B)\bar{C}$
C. $A \oplus B \oplus C$ **D.** $\bar{A}\bar{B}\bar{C}$

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Q2.24.1 **A**	Q2.23.1 **C**	Q2.22.1 **B**
Q2.20.1 **C**	Q2.18.1 **B**	Q2.12.1 **A**
Q2.11.1 **A**	Q2.09.1 **D**	Q2.08.1 **A**
Q2.07.1 **D**	Q2.06.1 **C**	Q2.05.1 **C**
Q2.04.1 **D**	Q2.03.1 **B**	Q2.00.1 **A**
Q2.98.1 **B**	Q2.97.1 **MTA**	Q2.95.1 **C**

PrepFusion

UNIT

III

Number Systems

Topics

1. Introduction to Number Systems
2. Arithmetic Operations on Numbers
3. Binary Codes and Their Operations
4. Signed Number Representations
5. Complement Arithmetic
6. Floating-Point Representation

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 10	**MCQ:** 9	**MSQ:** 0	**NAT:** 1
1M: 8	**2M:** 2	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2025)–1M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q3.25.1 MCQ 2025 1M

The 4-bit signed 2's complement form of $(5)_{10} + (5)_{10}$ is:

A. $(-6)_{10}$ **B.** $(-7)_{10}$ **C.** $(-5)_{10}$ **D.** $(-8)_{10}$

Q3.18.1 MCQ 2018 1M

The representation of the decimal number $(27.625)_{10}$ in base-2 number system is:

A. 11011.110 **B.** 11101.101
C. 11011.101 **D.** 10111.110

Q3.11.1 NAT 2011 1M

The base of the number system for the addition operation 24 + 14 = 41 is ____ .

Q3.09.1 MCQ 2009 1M

The binary representation of the decimal number 1.375 is

A. 1.111 **B.** 1.010 **C.** 1.011 **D.** 1.001

Q3.08.1 MCQ 2008 1M

The result of $(45)_{10} - (45)_{16}$ expressed in 2's complement representation is:

A. 011000 **B.** 100111 **C.** 101000 **D.** 101001

Q3.06.1 MCQ 2006 1M

A number N is stored in a 4-bit 2's complement representation as

a_3	a_2	a_1	a_0

It is copied into a 6-bit register and after a few operations the final bit pattern is

a_3	a_3	a_2	a_1	a_0	1

The value of the bit pattern in 2's complement representation is given in terms of the original number N as

A. $32a_3 + 2N + 1$ **B.** $32a_3 - 2N + 1$
C. $2N - 1$ **D.** $2N + 1$

Q3.05.1 MCQ 2005 1M

An 8-bit 2's complement representation of an integer is FA (Hex). Its decimal equivalent is:

A. -10 **B.** -6 **C.** +6 **D.** +10

Q3.04.1 MCQ 2004 1M

7E H and 5F H are XORed. The output is multiplied by 10 H. The result is:

A. 0210 H **B.** 7E5F H **C.** 5F7E H **D.** 2100 H

Q3.03.1 MCQ 2003 2M

Two 4-bit 2's complement numbers 1011 and 0110 are added. The result expressed in 4-bit 2's complement notation is:

A. 0001

B. 0010

C. 1101

D. cannot be expressed in 4-bit 2's complement

Q3.99.1 MCQ 1999 2M

A number in 4-bit 2's complement representation is $X_3X_2X_1X_0$. This number, when stored using 8 bits is:

A. $0000X_3X_2X_1X_0$ **B.** $1111X_3X_2X_1X_0$
C. $X_3X_3X_3X_3X_3X_2X_1X_0$ **D.** $\bar{X}_3\bar{X}_3\bar{X}_3\bar{X}_3X_3X_2X_1X_0$

Answer Key

Q3.25.1 **A**	Q3.18.1 **C**	Q3.11.1 **7**
Q3.09.1 **C**	Q3.08.1 **C**	Q3.06.1 **D**
Q3.05.1 **B**	Q3.04.1 **A**	Q3.03.1 **A**
Q3.99.1 **C**		

UNIT

IV

Combinational Circuits

Topics

1. Introduction to Combinational Circuits
2. Code Converters and 7-Segment Display
3. Parity Generator and Checker
4. Magnitude Comparators
5. Multiplexers (MUX) and Demultiplexers (DEMUX)
6. Implementing Logical Expressions Using MUX
7. Decoders, Encoders and Priority Encoders
8. Half Adder and Full Adder
9. Delay-Based Problems in Adders
10. Half Subtractor and Full Subtractor
11. Binary Parallel Adder with Delay-Based Problems
12. Look-Ahead Carry Adder
13. Binary Parallel Subtractor
14. Adder and Subtractor Circuits of Different Binary Codes
15. Binary Multiplier

Unit Snapshot*

Stream: IN	**Questions:** 16	**MCQ:** 14	**MSQ:** 1	**NAT:** 1
1M: 12	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2016)–3M(2019)

**See preface for how Avg Weightage and Range are calculated.*

Q4.25.1 MCQ 2025 2M

The minimized expression of the Boolean function $Y(P, Q, R)$ implemented by the multiplexer circuit shown in the figure is

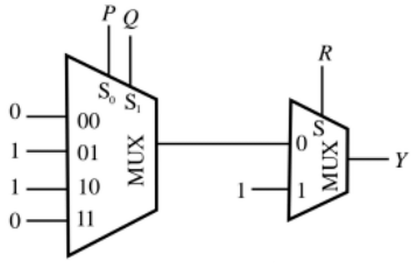


Fig. Q4.25.1

A. $Y = R + (P \oplus Q)$
B. $Y = R(P \oplus Q)$
C. $Y = R + \overline{(P \oplus Q)}$
D. $Y = R \oplus (P \oplus Q)$

Q4.22.1 MSQ 2022 2M

The output F of the digital circuit shown can be written in the form(s)

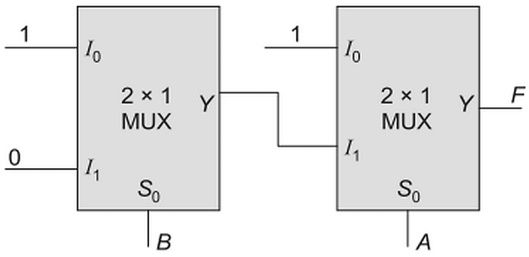


Fig. Q4.22.1

A. $\bar{A}.\bar{B}$
B. $\bar{A} + \bar{B}$
C. $\overline{A + B}$
D. $\overline{A.B}$

Q4.21.1 MCQ 2021 1M

A 4x1 MUX with two selection lines is used to realize Boolean function F having 4 Boolean variables x, y, z and w as shown below. S_0 and S_1 denote the least significant bit (LSB) and the most significant bit (MSB) of the selcector lines of multiplexer respectively. I_0 , I_1 , I_2 , I_3 are input lines of the multiplexer.

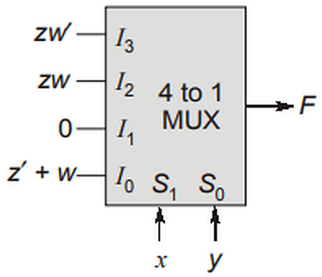


Fig. Q4.21.1

The canonical sum-of-product representation of F is:

A. $F(x, y, z, w) = \Sigma m(2, 5, 9, 11, 14)$

B. $F(x, y, z, w) = \Sigma m(0, 1, 3, 14, 15)$

C. $F(x, y, z, w) = \Sigma m(0, 1, 3, 11, 14)$

D. $F(x, y, z, w) = \Sigma m(1, 3, 7, 9, 15)$

Q4.21.2 MCQ 2021 1M

The Boolean operation performed by the following circuit at the output O is ____

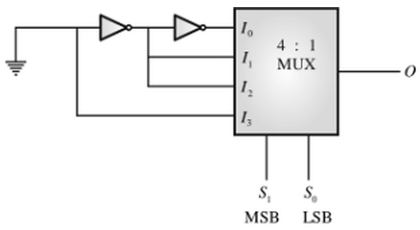


Fig. Q4.21.2

A. $O = S_1 + S_0$
B. $O = S_1 \oplus S_0$
C. $O = S_0.\bar{S}_1$
D. $O = S_1.\bar{S}_0$

Q4.19.1 MCQ 2019 1M

In the circuit shown below, assume that the comparators are ideal and all components have zero propagation delay. In one period of the input signal $V_{in} = 6\sin(\omega t)$, the fraction of the time for which the output OUT is in logic state HIGH is:

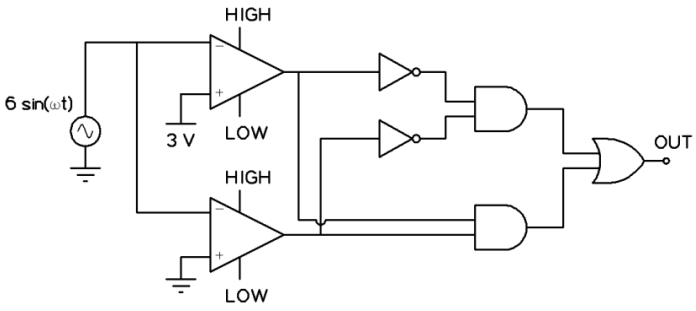


Fig. Q4.19.1

A. $\frac{1}{12}$ **B.** $\frac{1}{2}$ **C.** $\frac{2}{3}$ **D.** $\frac{5}{6}$

Q4.19.2 NAT 2019 2M

The given figure shows the ith full-adder bock of a binary adder circuit. C_i is the input carry and C_{i+1} is the output carry of the circuit. Assume that each logic gate has a delay of 2ns, with no additional delay due to the interconnecting wires. If the inputs A_i, B_i are available and stable throughout the carry propagation, the maximum time taken for an input C_i to produce a steady-state output C_{i+1} is ____ ns .

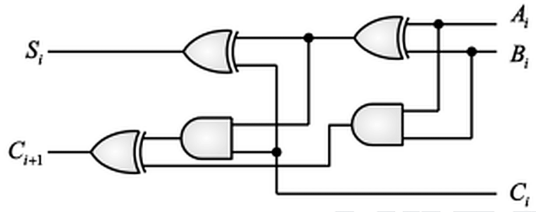


Fig. Q4.19.2

Q4.16.1 MCQ 2016 1M

A 4x1 MUX to realize a Boolean function $F(X, Y, Z)$ is shown in the figure below. The inputs Y and Z are connected to the selectors of the MUX (Y is more significant). The canonical sum-of-product expression for $F(X, Y, Z)$ is:

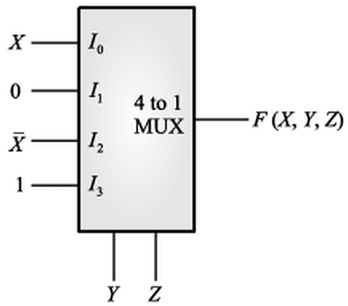


Fig. Q4.16.1

A. $\Sigma m(2, 3, 4, 7)$ **B.** $\Sigma m(1, 3, 5, 7)$
C. $\Sigma m(0, 2, 4, 6)$ **D.** $\Sigma m(2, 3, 5, 6)$

Q4.12.1 MCQ 2012 1M

The output Y of a 2-bit comparator is logic 1 whenever the 2-bit input A is greater than the 2-bit input B . The number of combinations for which the output is logic 1, is:

A. 4 **B.** 6 **C.** 8 **D.** 10

Q4.08.1 MCQ 2008 1M

The output F of the multiplexer circuit shown below expressed int erms of the inputs P, Q and R is:

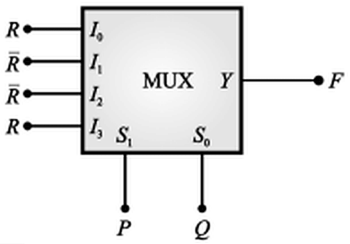


Fig. Q4.08.1

A. $F = P \oplus Q \oplus R$ **B.** $F = PQ + QR + RP$
C. $F = (P \oplus Q)R$ **D.** $F = (P \oplus Q)\bar{R}$

Q4.07.1 MCQ 2007 1M

A MUX circuit shown in the figure below implements a logic function. The correct expression for F_1 is:

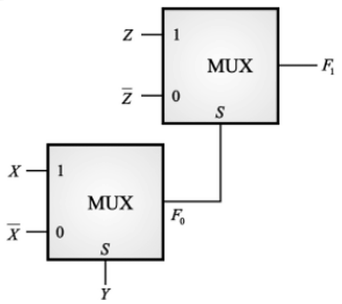


Fig. Q4.07.1

A. $\overline{(X \oplus Y)} \oplus Z$ **B.** $\overline{\overline{(X \oplus Y)} \oplus Z}$
C. $(X \oplus Y) \oplus \bar{Z}$ **D.** $(X \oplus Y) \oplus Z$

Q4.06.1 MCQ 2006 1M

A combinational circuit using an 8-to-1 multiplexer is shown in the following figure. The minimized expression for the output (Z) is:

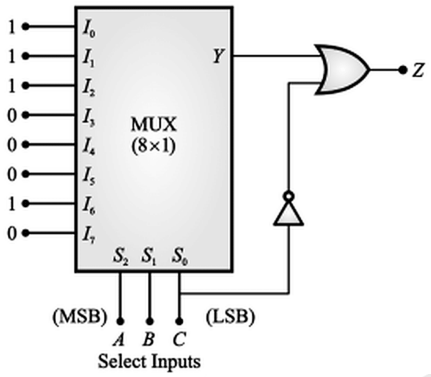


Fig. Q4.06.1

A. $C(\bar{A}+\bar{B})$
B. $C(A+B)$
C. $\bar{C}+\bar{A}\bar{B}$
D. $\bar{C}+AB$

Q4.05.1 MCQ 2005 1M

The cell of a field programmable Gate array is shown in the figure. It has three 2x1 multiplexers with their select lines G_0, G_1, G_2 and 4 digital signal input lines I_0, I_1, I_2, I_3. The logical function that relates the output O to the select and signal input lines is:

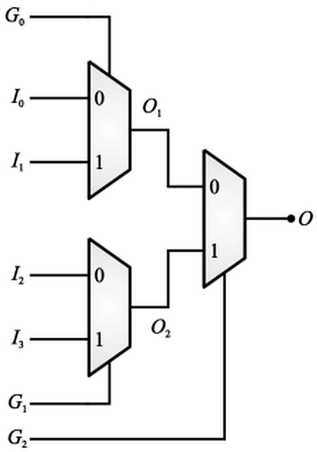


Fig. Q4.05.1

A. $\bar{G}_0\bar{G}_1I_2+\bar{G}_0\bar{G}_1I_3+\bar{G}_2\bar{G}_1I_0+\bar{G}_2\bar{G}_1I_1$

B. $\bar{G}_0I_2+\bar{G}_0G_1+\bar{G}_2I_0+\bar{G}_2G_1I_1+G_0$

C. $\bar{G}_0\bar{G}_2I_0+G_0\bar{G}_2I_1+G_2\bar{G}_1I_2+G_2G_1I_3$

D. $G_2G_1\bar{I}_2+\bar{G}_2\bar{G}_1I_3+G_2\bar{G}_0I_0+G_0\bar{G}_2I_1$

Q4.04.1 MCQ 2004 1M

In the given circuit , A, B, C are the inputs and P, Q are the two outputs.

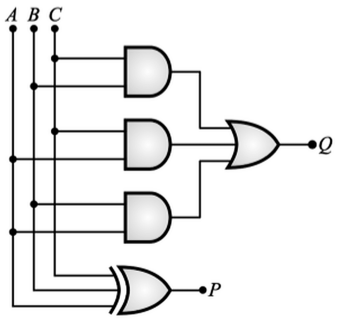


Fig. Q4.04.1

The circuit is a:

A. Half adder where P is the sum and Q is the carry

B. Half adder where P is the carry and Q is the sum

C. Full adder where P is the sum and Q is the carry

D. Full adder where P is the carry and Q is the sum

Q4.03.1 MCQ 2003 1M

The logic circuit given below is:

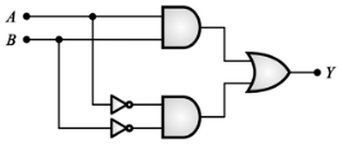


Fig. Q4.03.1

A. Half adder
B. XOR
C. Equality detector
D. Full adder

Q4.03.2 MCQ 2003 1M

A 2X1 digital multiplexer having a switching delay of 1 μs is connected as shown in figure. The output of the multiplecer is tied to its own select input S . The inputs which get selected when $S = 0$ is tied to 1, and the input that gets selected when $S = 1$ is tied to 0. The output V_0 will be

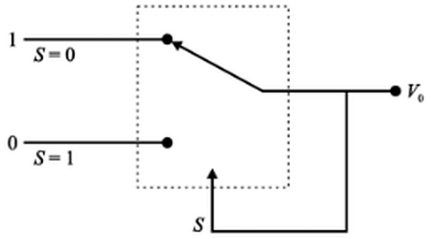


Fig. Q4.03.2

A. 0

B. 1

C. pulse train of frequency 0.5 MHz

D. pulse train of frequency 1.0 MHz

Q4.95.1 MCQ 1995 2M

The combinatorial circuit shown in figure employs a 4x1 multiplexer. The output Q of the circuit is:

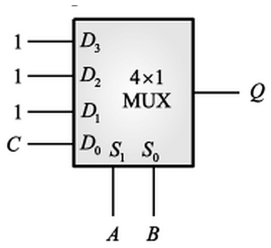


Fig. Q4.95.1

A. $\bar{A}\bar{B}C$

B. $A + B + C$

C. $A \oplus B \oplus C$

D. $\bar{A}\bar{B}\bar{C}$

Answer Key

Q4.25.1 A	Q4.22.1 B , D	Q4.21.1 C
Q4.21.2 B	Q4.19.1 D	Q4.19.2 4
Q4.16.1 A	Q4.12.1 B	Q4.08.1 A
Q4.07.1 D	Q4.06.1 C	Q4.05.1 C
Q4.04.1 C	Q4.03.1 C	Q4.03.2 C
Q4.95.1 B		

UNIT

V

Sequential Circuits

Topics

1. Latches — SR, JK, D and T Types, Their Drawbacks and Gated Latches
2. Flip-Flops — Introduction, Types, Conversions, Race Around Condition and Master-Slave Configuration
3. Asynchronous Counters — Binary, Non-Binary (Up/Down) and Output Decoding
4. Synchronous Counters — Propagation Delay, Setup/Hold Time, Critical Path Delay, Clear and Preset Input
5. Shift Registers and Applications — Ring Counter, Johnson Counter and Cascading
6. FSM and Sequence Detectors — Design, Generation, Detection and Overlapping/Non-Overlapping Types
7. Static Timing Analysis — Setup/Hold Time, Clock Constraints, Critical Path and Timing Violations

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 43	**MCQ:** 41	**MSQ:** 0	**NAT:** 2
1M: 39	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2022)–3M(2012)

**See preface for how Avg Weightage and Range are calculated.*

Q5.26.1 MCQ 2026 2M

The initial state $Q_3Q_2Q_1Q_0$ in the digital circuit shown below is 1011. The new value of $Q_3Q_2Q_1Q_0$ after two rising edges of the clock (CLK) will be ____ .

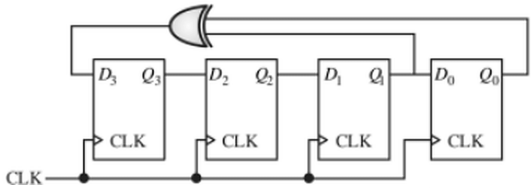


Fig. Q5.26.1

A. 1101 **B.** 0101 **C.** 1010 **D.** 1011

Q5.25.1 MCQ 2025 2M

The clock frequency of the digital circuit shown in the figure is 12 MHz. The frequencies of the output (F) corresponding to Control=0 and Control=1, respectively, are:

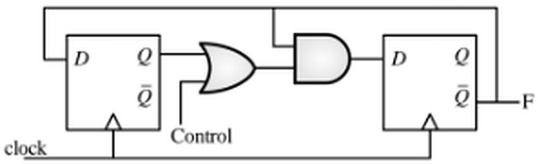


Fig. Q5.25.1

A. 4 MHz and 6 MHz **B.** 6 MHz and 4 MHz
C. 3 MHz and 4 MHz **D.** 3 MHz and 6 MHz

Q5.24.1 MCQ 2024 2M

In the figure shown,the positive edge-triggered D flip-flops are initially reset to $Q = 0$. The logic gates and the multiplexers have no propagation delay. After reset, a train of clock pulses (CLK) are applied. The logic states of the inputs DIN , S and the clock pulses are also shown in the figure. Assuming no timing violations, the sequence of output Y from the 3rd clock to the 5th clock , $Y_3Y_4Y_5$ is

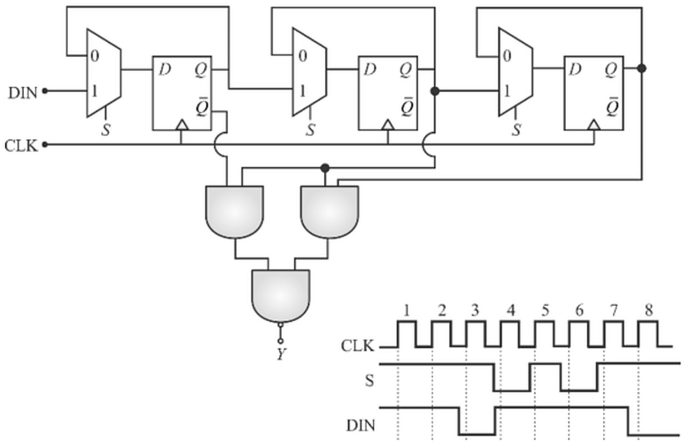


Fig. Q5.24.1

A. 001 **B.** 010 **C.** 000 **D.** 011

Q5.23.1 MCQ 2023 1M

In the circuit shown, the initial binary content of shift register A is 1101 and that of shift register B is 1010. The shift registers are positive edge triggered and the gates have no delay. When the shift control is high, what will be the binary content of the shift registers A and B after 4 clock pulses?

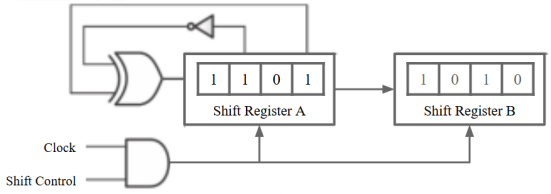


Fig. Q5.23.1

A. A = 1101 , B = 1101 **B.** A = 1110 , B = 1001
C. A = 0101 , B = 1101 **D.** A = 1010 , B =1111

Q5.23.2 MCQ 2023 1M

The table shows present state $Q(t)$, next state $Q(t+1)$, and the control input in a flip flop. Identify the flip-flop.

$Q(t)$	$Q(t+1)$	Input
0	0	0
0	1	1
1	0	1
1	1	0

A. T flip-flop **B.** D flip-flop
C. SR flip-flop **D.** JK flip-flop

Q5.22.1 MCQ 2022 1M

The digital circuit shown

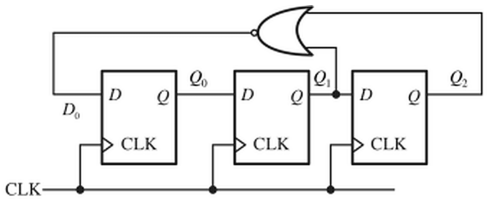


Fig. Q5.22.1

A. is a divide-by-8 counter

B. is a divide-by-7 counter

C. is a divide-by-5 counter

D. does not function as a counter due to disjoint cycles of states

Q5.21.1 MCQ 2021 1M

Given below is the diagram of a synchronous sequential circuit with one JK flip-flop and one T flip-flop with their outputs denoted as A and B respectively, with $J_A = \bar{A} + \bar{B}$ and $K_A = A + B$ and $T_B = A$.

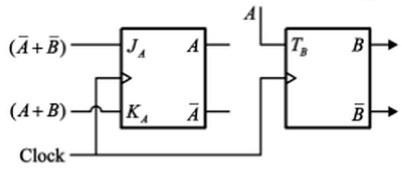


Fig. Q5.21.1

Starting from the initial state $(AB = 00)$, the sequence of states (AB) visited by the circuit is

A. 00 → 10 → 11 → 01 → 00...

B. 00 → 01 → 10 → 11 → 00...

C. 00 → 10 → 01 → 11 → 00...

D. 00 → 01 → 11 → 00...

Q5.20.1 NAT 2020 1M

Two T-flipflops are interconnected as shown in the figure. The present state of the flipflops are : $A = 1, B = 1$. The input X is given as 1 , 0 , 1 in the next three clock cycles. The decimal equivalent of $(ABY)_2$ with A being the MSB and Y being the LSB, after the 3rd clock cycle is ____ .

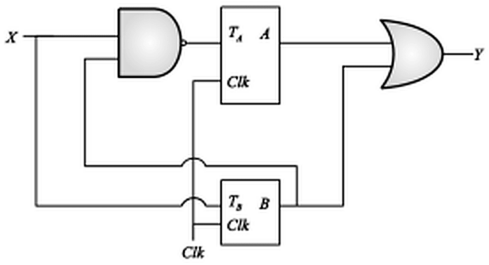


Fig. Q5.20.1

Q5.19.1 MCQ 2019 1M

The circuit shown in the figure below uses ideal positive edge-triggered synchronous JK flip-flops with outputs X and Y. If the initial state of the output is X=0 and Y=0 just before the arrival of the first clock pulse, the state of the output just before the arrival of the second clock pulse is:

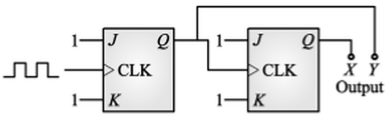


Fig. Q5.19.1

A. X = 0 , Y = 0 **B.** X = 0 , Y = 1
C. X = 1 , Y = 0 **D.** X = 1 , Y = 1

Q5.18.1 MCQ 2018 1M

A 2-bit synchronous counter using two $J - K$ flip-flops is shown. The expressions for the inputs to the $J-K$ flip-flops are also shown in the figure. The output sequence of the counter starting from $Q_1Q_2 = 00$ IS:

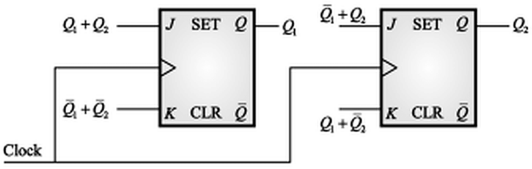


Fig. Q5.18.1

A. 00 → 11 → 10 → 01 → 00...

B. 00 → 01 → 10 → 11 → 00...

C. 00 → 01 → 11 → 10 → 00...

D. 00 → 10 → 11 → 01 → 00...

Q5.18.2 NAT 2018 1M

For the 3-bit binary counter shown in the figure, the output increments at every positive transition in the clock (CLK) . Assume ideal diodes and the starting state of the counter as 000. If output high is 1V and output low is 0V, the current I (in mA) flowing through the 50 Ω resistor during the 5th clock cycle is (up to one decimal place) .

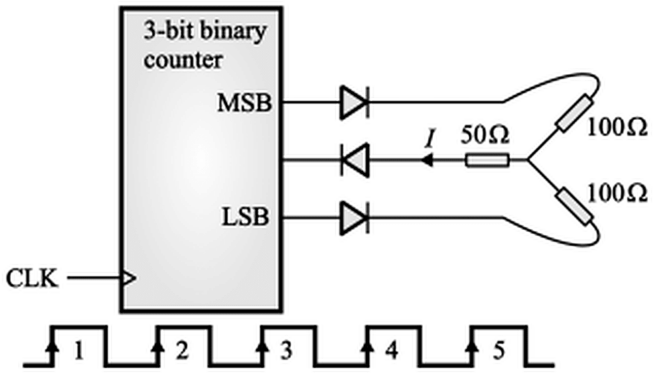


Fig. Q5.18.2

Q5.17.1 MCQ 2017 1M

The two inputs A and B are connected to an R-S latch via two AND gates as shown in the figure. If $A = 1$ and $B = 0$, the output $Q\bar{Q}$ is:

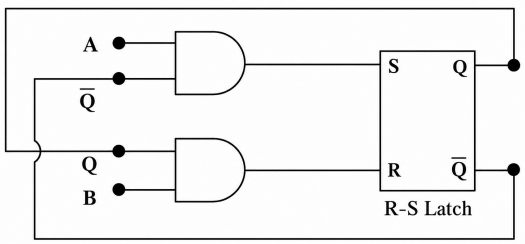


Fig. Q5.17.1

A. 00 **B.** 10 **C.** 01 **D.** 11

Q5.16.1 MCQ 2016 1M

A synchronous counter using two J-K flipflops that goes through the sequence of states : $Q_1Q_2 = 00 \rightarrow 10 \rightarrow 01 \rightarrow 11 \rightarrow 00$... is required. To achieve this, the inputs to the flipflops are:

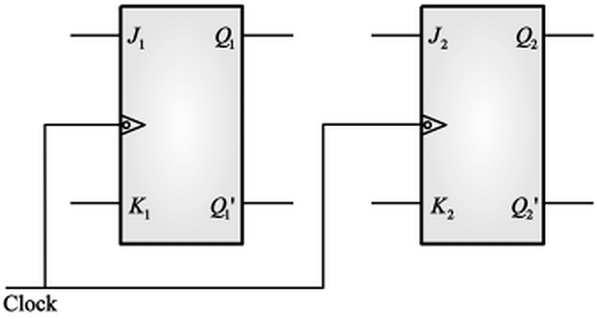


Fig. Q5.16.1

A. $J_1 = Q_2$, $K_1 = 0$, $J_2 = Q_1'$, $K_2 = Q_1$

B. $J_1 = 1$, $K_1 = 1$, $J_2 = Q_1$, $K_2 = Q_1$

C. $J_1 = Q_2$, $K_1 = Q_2'$, $J_2 = Q_2$, $K_2 = 1$

D. $J_1 = Q_2'$, $K_1 = Q_2$, $J_2 = Q_1'$, $K_2 = Q_1'$

Q5.15.1 MCQ 2015 1M

The number of clock cycles for the duration of an input pulse is counted using a cascade of N decade counters (DC 1 to DC N) as shown in the figure. If the clock frequency in MHz is f , the resolution and range of measurement of input pulse width , both in μs , are respectively,

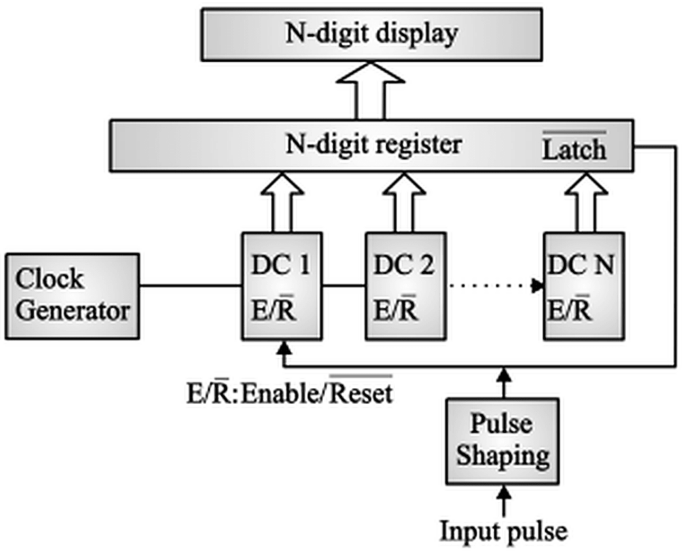


Fig. Q5.15.1

A. $\frac{1}{f}$ and $\frac{(2^N - 1)}{f}$ **B.** $\frac{1}{f}$ and $\frac{(10^N - 1)}{f}$

C. $\frac{10^N}{f}$ and $\frac{(10^N - 1)}{f}$ **D.** $\frac{2^N}{f}$ and $\frac{(2^N - 1)}{f}$

Q5.15.2 MCQ 2015 1M

In the circuit shown, the switch is momentarily closed and then opened. Assuming the logic gate to have equal non-zero delay , at steady state , the logic states of X and Y are

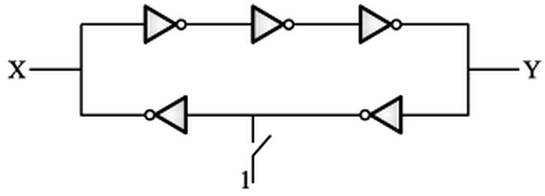


Fig. Q5.15.2

A. X is latched , Y toggles continuously

B. X and Y are both latched

C. Y is latched , X toggles continuously

D. X and Y both toggle continuously

Q5.14.1 MCQ 2014 1M

Frequency of an analog periodic signal in the range of 5 kHz - 10 kHz is to be measured with a resolution of 100 Hz by measuring its period with a counter. Assuming negligible signal and transition delays the minimum clock frequency and minimum number of bits in the counter needed , respectively , are:

A. 1 MHz , 10 bits **B.** 10 MHz , 10 bits
C. 1 MHz , 8 bits **D.** 10 MHz , 8 bits

Q5.13.1 MCQ 2013 1M

The digital circuit shown below uses two negative edge-triggered D flip-flops. Assuming initial condition of Q_1 and Q_0 as zero, the output Q_1Q_0 of this circuit is

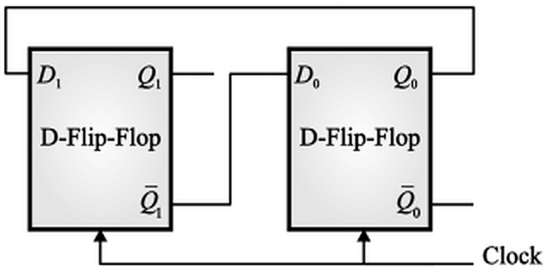


Fig. Q5.13.1

A. 00 → 01 → 10 → 11 → 00...

B. 00 → 01 → 11 → 10 → 00...

C. 00 → 11 → 10 → 01 → 00...

D. 00 → 01 → 11 → 11 → 00...

Q5.12.1 MCQ 2012 2M

The state transition diagram for the logic circuit shown is

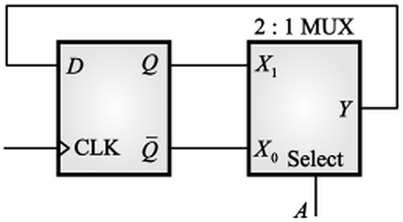
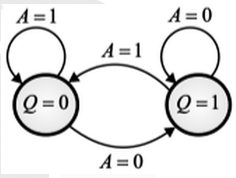
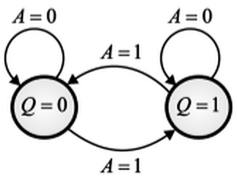
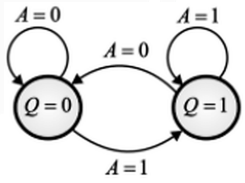
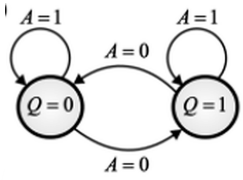


Fig. Q5.12.1

A.


B.


C.


D.


Q5.12.2 MCQ 2012 1M

Consider the given circuit.

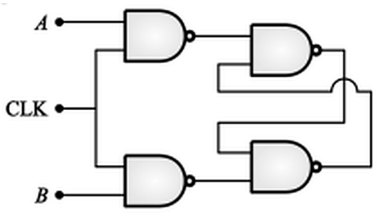


Fig. Q5.12.2

In this circuit, the race around condition

A. does not occur

B. occurs when CLK=0

C. occurs when CLK=1 and $A = B = 1$

D. occurs when CLK=1 and $A = B = 0$

Q5.11.1 MCQ 2011 1M

The circuit below shows an up/down counter working with a decoder and a flip-flop. Preset and clear of the flip-flop are asynchronous active-low inputs.

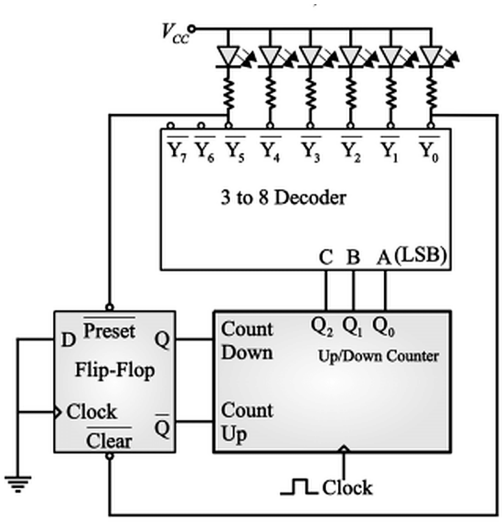


Fig. Q5.11.1

Assuming that the initial value of counter output ($Q_2Q_1Q_0$) as zero, the counter outputs in decimal for 12 clock cycles are

A. 0 , 1 , 2 , 3 , 4 , 4 , 3 , 2 , 1 , 1 , 2 , 3 , 4

B. 0 , 1 , 2 , 3 , 4 , 5 , 0 , 1 , 2 , 3 , 4 , 5 , 0

C. 0 , 1 , 2 , 3 , 4 , 5 , 5 , 4 , 3 , 2 , 1 , 0 , 1

D. 0 , 1 , 2 , 3 , 4 , 5 , 4 , 3 , 2 , 1 , 0 , 1 , 2

Q5.09.1 MCQ 2009 1M

In the figure shown , the initial state Q is 0. The output is observed after the application of each clock pulse. The output sequence at Q is:

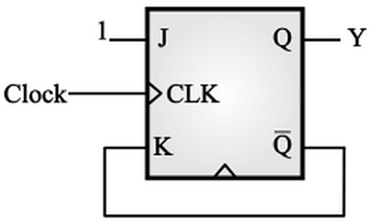


Fig. Q5.09.1

A. 0000.... **B.** 1010.... **C.** 1111.... **D.** 1000...

Q5.09.2 MCQ 2009 1M

The given figure shows a 3-bit ripple counter, with Q_2 as MSB. The flip-flops are rising edge triggered. The counting direction is:

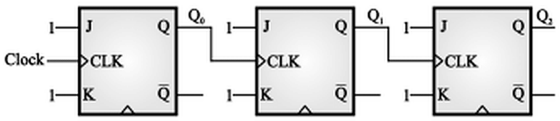


Fig. Q5.09.2

A. always down

B. always up

C. up or down depending on the initial state of Q_0 only

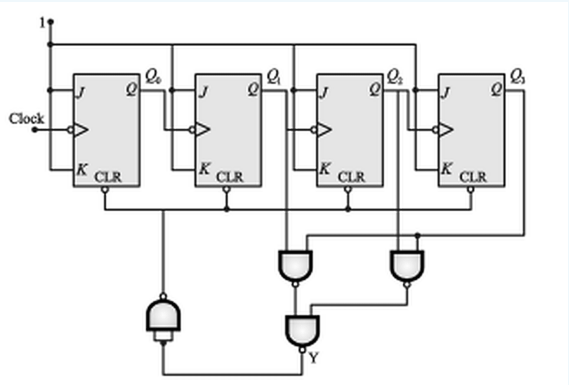
D. up or down depending on the initial states of Q_0, Q_1, Q_2

Common Data Questions

The next 2 questions share the data below.

Answer the next two questions using the following data.

Consider the counter circuit shown below,



Q5.08.1 MCQ 2008 1M

In the given figure, Y can be expressed as:

A. $Q_3(Q_1 + Q_2)$ **B.** $Q_3 + Q_1Q_2$

C. $\overline{Q_3(Q_1 + Q_2)}$ **D.** $\overline{Q_3 + Q_1Q_2}$

Q5.08.2 MCQ 2008 1M

The given circuit is a

A. Mod-8 Counter **B.** Mod-9 Counter
C. Mod-10 Counter **D.** Mod-11 Counter

Q5.07.1 MCQ 2007 1M

The pulse width T of an asynchronous pulse is measured by a counter with an edge-triggered clock of known frequency f_c as shown in the figure below:

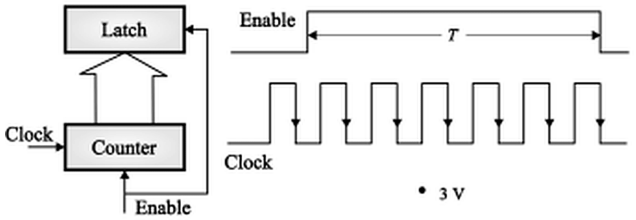


Fig. Q5.07.1

The pulse, whose width is to be measured, is applied to the Enable pin of the counter. The counter counts while the Enable is high and is held reset to zero otherwise. The counter output is latched by the negative edge of the Enable signal. The measured pulse width is taken to be N times the clock period, where N is the count reach at the end of a count cycle. Assuming no overflow, the measurement error will be limited to x% of T if

A. $T > \dfrac{100}{x f_c}$ **B.** $T < \dfrac{100}{x f_c}$

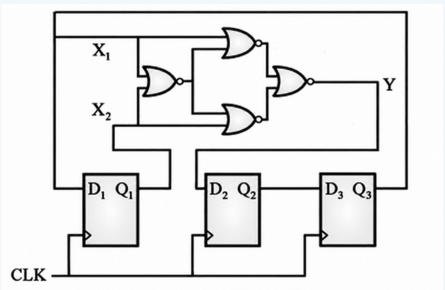
C. $T > \dfrac{200}{x f_c}$ **D.** $T < \dfrac{200}{x f_c}$

Common Data Questions

The next 2 questions share the data below.

Answer the next two questions using the following data.

Consider the counter circuit shown in the following figure:



Q5.07.2 MCQ 2007 1M

The correct input-output relationship between Y and X_1X_2 is:

A. $Y = X_1 + X_2$ **B.** $Y = X_1X_2$

C. $Y = X_1 \oplus X_2$ **D.** $Y = \overline{X_1 \oplus X_2}$

Q5.07.3 MCQ 2007 1M

The D flipflops are initialized to $Q_1Q_2Q_3 = 000$. After 1 clock cycle, $Q_1Q_2Q_3$ is equal to

A. 001 **B.** 010 **C.** 100 **D.** 101

Q5.07.4 MCQ 2007 1M

The square wave C_1 shown in the figure is given to the clock input of a 4-bit binary up/down conter whose UP/$\overline{\text{DN}}$ input is fed with the train p_v. The counter is a negative edge-triggered one. The counter starts with 0000 and will reach 0000 again at the

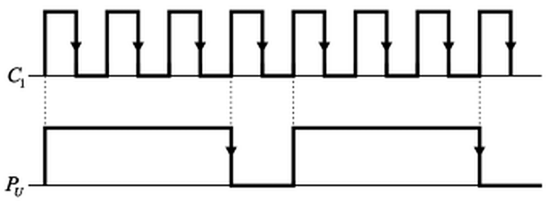


Fig. Q5.07.4

A. 15th clock pulse **B.** 16th clock pulse

C. 44th clock pulse **D.** 48th clock pulse

Q5.07.5 MCQ 2007 1M

A sequential circuit is shown in the given figure. Let the state of the circuit be encoded as Q_A, Q_B . The notation X $\rightarrow$ Y implies that state Y is reachable from state X in a finite number of clock transitions.

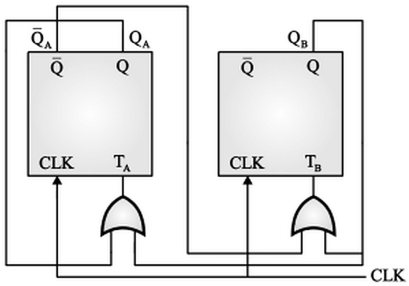


Fig. Q5.07.5

Identify the INCORRECT statement.

A. 01 → 00 **B.** 11 → 01 **C.** 01 → 11 **D.** 01 → 10

Q5.06.1 MCQ 2006 1M

All the logic gates in the circuit shown have finite propagation delay. The circuit can be used as a clock generator, if

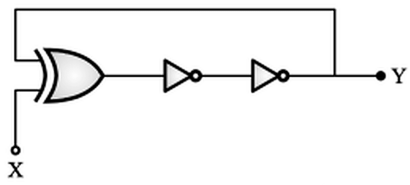


Fig. Q5.06.1

A. X=0 **B.** X=1 **C.** X=0 or 1 **D.** X=Y

Q5.06.2 MCQ 2006 1M

Given that the initial state (Q_1Q_0) is 00 , the counting sequence of the counter shown in the following figure for Q_1Q_0 is

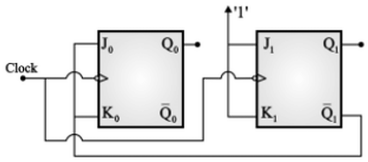


Fig. Q5.06.2

A. 00 - 11 - 01 - 10 - 00

B. 00 - 01 - 11 - 10 - 00

C. 00 - 11 - 10 - 01 - 00

D. 00 - 10 - 01 - 11 - 00

Q5.05.1 MCQ 2005 1M

The two NAND gates before latch circuit shown in the figure are used to

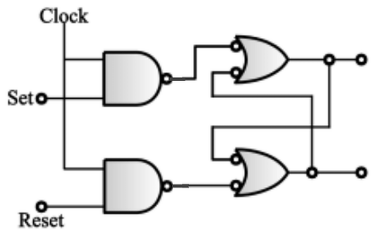


Fig. Q5.05.1

A. act as buffers

B. operate the latch faster

C. avoid racing problem

D. invert the latching action

Q5.05.2 MCQ 2005 1M

For the digital counter shown in the figure with outputs Q_0, Q_1, Q_2 ... where Q_0 indicates the LSB of the count value, the correct statement when the switch Sw is closed

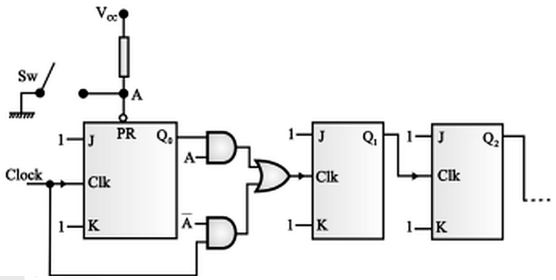


Fig. Q5.05.2

A. Counter outputs are both even and odd numbers

B. Counter outputs are only odd numbers

C. Counter outputs are only even numbers

D. Counter stops counting

Q5.04.1 MCQ 2004 1M

In the digital circuit shown in the figure, the flipflops have set time of 5 ns and a worst case delay of 15 ns. The AND gate has a delay of 5 ns. maximum possible clock rate for the circuit to operate faithfully is

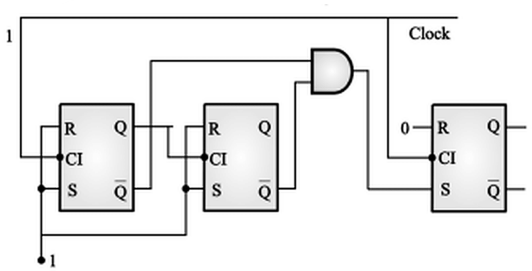


Fig. Q5.04.1

A. 21 MHz **B.** 22 MHz **C.** 25 MHz **D.** 30 MHz

Q5.04.2 MCQ 2004 1M

In the figure, initially $Q = A = B = 0$. After three clock triggers, the state of Q, A and B will be respectively

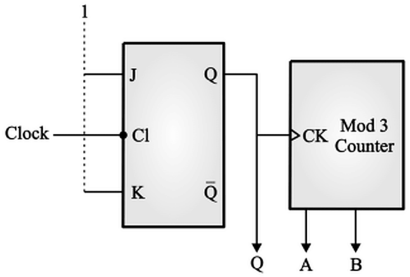


Fig. Q5.04.2

A. 110 **B.** 011 **C.** 001 **D.** 101

Q5.03.1 MCQ 2003 1M

The correct cyclic sequence of the outputs Q_0Q_1 for the master-slave JK flipflop circuit shown in below figure, when the input clock pulse applied is:

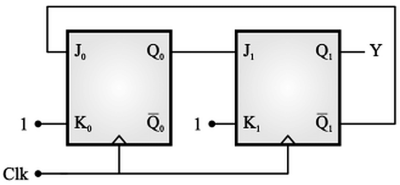


Fig. Q5.03.1

A. 00 , 01 , 10 , 11 , 00 , 01 , ...

B. 00 , 10 , 01 , 00 , 10 , 01 , ...

C. 00 , 01 , 10 , 00 , 01 , 10 , ...

D. 00 , 11 , 00 , 11 , 00 , 11 , ...

Q5.98.1 MCQ 1998 1M

The minimum number of flipflops needed to make mod-2 counter is

A. 1 **B.** 2 **C.** 3 **D.** 4

Q5.97.1 MCQ 1997 1M

A 4-bit synchronous counter with a series carry, uses flipflop and AND gates, having a propagation delay of 30 ns and 10 ns respectively. The maximum time interval required between two successive clock pulses for reliable operation of the counter is

A. 10 ns **B.** 30 ns **C.** 40 ns **D.** 50 ns

Q5.95.1 MCQ 1995 1M

Figure shows a sequential circuit with four JK flipflops and generate a table of output $Q_3Q_2Q_1Q_0$ changes with each clock pulse. Start with $Q_3Q_2Q_1Q_0 = 0001$ and the next state after the first clock pulse

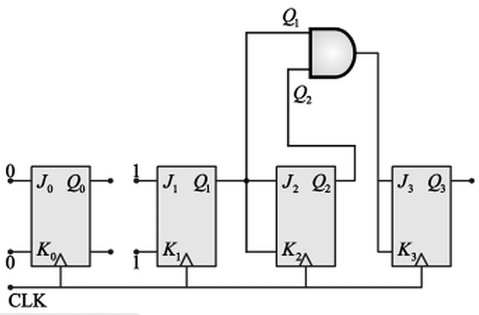


Fig. Q5.95.1

A. 1111 **B.** 1100 **C.** 0001 **D.** 0011

Q5.95.2 MCQ 1995 1M

The circuit shown in figure is a

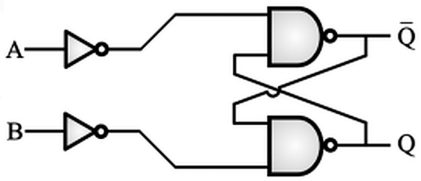


Fig. Q5.95.2

A. R-S latch **B.** R-S flipflop
C. D latch **D.** D flipflop

Q5.94.1 MCQ 1994 1M

The minimum number of flipflops required to design a MOD-10 counter is

A. 3 **B.** 10 **C.** 4 **D.** 5

Q5.94.2 MCQ 1994 1M

The output Q_{n+1} of a JK flipflop for the input $J = 1, K = 1$ is

A. 0 **B.** 1 **C.** Q_n **D.** $\bar{Q}_n$

Q5.98.1	Q5.97.1	Q5.95.1
A	C	D
Q5.95.2	Q5.94.1	Q5.94.2
A	C	D
Q5.93.1		
A		

Q5.93.1 MCQ 1993 1M

A sequential circuit formed by two edge triggered JK flipflops is shown in figure. If the states before the arrival of a clock edge are $Q_1 = 1, Q_2 = 0$,the states after the arrival of the clock edge are

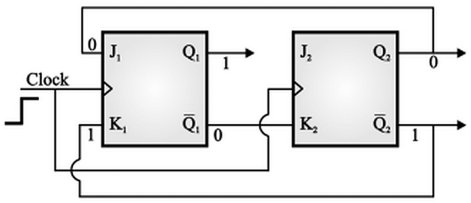


Fig. Q5.93.1

A. $Q_1Q_2 = 01$
B. $Q_1Q_2 = 11$
C. $Q_1Q_2 = 10$
D. $Q_1Q_2 = 00$

Answer Key

Q5.26.1	Q5.25.1	Q5.24.1
C	A	A
Q5.23.1	Q5.23.2	Q5.22.1
C	A	C
Q5.21.1	Q5.20.1	Q5.19.1
C	7	D
Q5.18.1	Q5.18.2	Q5.17.1
C	10	B
Q5.16.1	Q5.15.1	Q5.15.2
B	B	D
Q5.14.1	Q5.13.1	Q5.12.1
C	B	D
Q5.12.2	Q5.11.1	Q5.09.1
A	D	C
Q5.09.2	Q5.08.1	Q5.08.2
A	A	C
Q5.07.1	Q5.07.2	Q5.07.3
B	D	B
Q5.07.4	Q5.07.5	Q5.06.1
C	C	B
Q5.06.2	Q5.05.1	Q5.05.2
A	D	B
Q5.04.1	Q5.04.2	Q5.03.1
C	A	B

UNIT

VI

ADC & DAC

Topics

1. Introduction to DAC
   — Weighted Resistor and R-2R Ladder Type
2. Introduction to ADC
   — Counter Type
   — SAR Type ADC
   — Flash Type ADC
   — Single Slope and Dual Slope ADC
   — Sample and Hold Operation in ADC

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 39	**MCQ:** 28	**MSQ:** 1	**NAT:** 10
1M: 34	**2M:** 5	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2026)–4M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q6.26.1 NAT 2026 1M

The minimum number of comparators required in a 6-bit flash analog-to-digital converter is ____ . (answer in integer)

Q6.25.1 MCQ 2025 2M

A 3-bit DAC is implemented using ideal opamp and switches as shown in the figure.

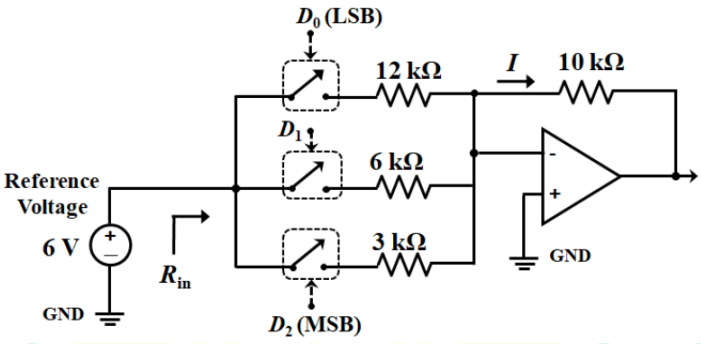


Fig. Q6.25.1

Each of the switches gets closed when its corresponding digital input is at logic 1. For a digital input of 110, the resistance R_{in} seen from the reference source and the current I, are

A. R_{in} = 2 k Ω and I = 3 mA

B. R_{in} = 12 k Ω and I = 0.5 mA

C. R_{in} = ∞ Ω and I = 1 mA

D. R_{in} = ∞ Ω and I = 3 mA

Q6.25.2 NAT 2025 2M

A dual-slope ADC has a fixed integration time of 100 ms. The reference voltage of the ADC is -5V. The time taken by the ADC to measure an input voltage of 1.25 V is ____ ms (rounded off to the nearest integer).

Q6.24.1 NAT 2024 1M

An ADC has a full-scale voltage of 1.4 V, resolution of 200 mV, and produces binary output data. The input signal od the ADC has a bandwidth of 500 MHz, and it samples the data at the Nyquist rate. The parallel data output is converted to a serial bit stream using a parallel-to-serial converter. The data rate at the output of the parallel-to-serial converter is ____ Gbps. (rounded off to nearest integer)

Q6.22.1 MCQ 2022 2M

A resistor ladder digital-to-analog converter (DAC) receives a digital input that results in the circuit having the state as shown in the figure. For this digital input, the Thevenin voltage V_{th}, and Thevenin resistance R_{th}, as seen at the output node are

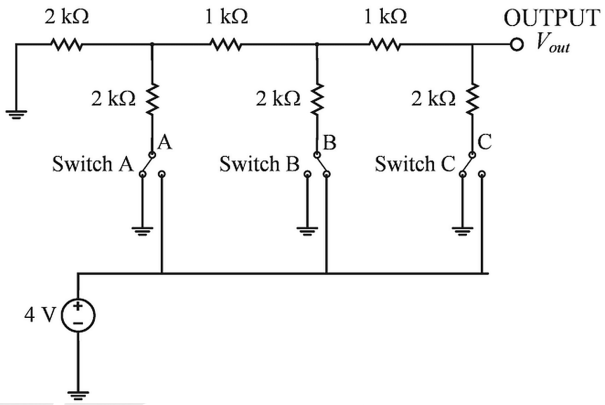


Fig. Q6.22.1

A. V_{th} = 1 V , R_{th} = 1 k Ω

B. V_{th} = 0.5 V , R_{th} = 2 k Ω

C. V_{th} = 0.5 V , R_{th} = 1 k Ω

D. V_{th} = 1 V , R_{th} = 2 k Ω

Q6.22.2 NAT 2022 1M

An analog-to-digital converter with resolution 0.01 V converts analog signals between 0 V to +10 V to an unsigned binary output. The minimum number of bits (in integer) in the output is ____.

Q6.21.1 NAT 2021 1M

A 10-bit ADC has a full-scale of 10.23 V , when the digital output is $(11\ 1111\ 1111)_2$. The quantization error of the ADC in millivolts is ____.

Q6.21.2 MCQ 2021 1M

A $10\frac{1}{2}$ digital counter-timer is set in the "frequency mode" of operation (with T_s = 1s) . For a specific input, the rating obtained is 1000. Without disconnecting this input, the counter-timer is changed to operate in the "Period mode" and the range selected is microseconds (μs , with f_s = 1 MHz). The counter will then display

A. 1000 **B.** 100 **C.** 0 **D.** 10

Q6.21.3 MSQ 2021 2M

For a 4-bit flash type Analog-to-Digital Converter(ADC) with full scale input voltage range "V", which of the following statement(s) is/are true?

A. A change in input voltage by $\frac{V}{16}$ will always flip MSB of the output.

B. A change in input voltage by $\frac{V}{16}$ will always flip LSB of the output.

C. The ADC requires one 4-to-2 priority encoder and 4 comparators.

D. The ADC requires 15 comparators.

Q6.20.1 NAT 2020 2M

The address lines A_9.....A_2 of 10 bit, 1.023 V full scale DAC is connected to the data lines D_7 to D_0 of an 8-bit microprocessor with A_1 and A_0 of the DAC grounded. Now, D_7.....D_0 is changed from 1010 1010 to 1010 1011. The corresponding change in the output of the DAC (in mV, rounded off to one decimal place) is ____ .

Q6.20.2 NAT 2020 1M

A 200 mV full scale dual slope analog-to-digital converter (DS-ADC) has a reference voltage of 100 mV. The first integration time is set as 100 ms. The DS-ADC is operated in the continuous conversion mode. The conversion time of the DS-ADC for an input voltage of 123.4 mV (in ms, rounded off to one decimal place) is ____ .

Q6.19.1 NAT 2019 1M

An 8-bit weighted resistor DAC has the smallest resistance of 500 Ω. The largest resistance has a value of ____ kΩ .

Q6.18.1 MCQ 2018 1M

The number of comparators required for implementing an 8-bit flash ADC is

A. 8 **B.** 128 **C.** 255 **D.** 256

Q6.14.1 MCQ 2014 1M

An N-bit ADC has an analog reference voltage V. Assuming zero mean and uniform distribution of the quantization error, the quantization noise power will be

A. $\frac{V^2}{12(2^N-1)^2}$ **B.** $\frac{V^2}{12(2^N-1)}$

C. $\frac{V}{12(2^N-1)}$ **D.** $\frac{V^2}{\sqrt{12}}$

Q6.14.2 MCQ 2014 1M

The circuit in the figure represents a counter-based unipolar ADC. When SOC is asserted the counter is reset and clock is enabled so that the counter counts up and the DAC output grows. When the DAC output exceeds the input sample value, the comparator switches from logic 0 to logic 1, disabling the clock and enabling the output buffer by asserting EOC. Assuming all components to be ideal, V_{ref} , DAC output and input to be positive, the maximum error in conversion of the analog sample value is:

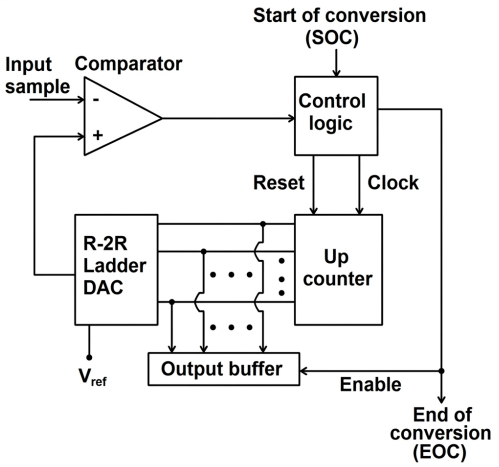


Fig. Q6.14.2

A. directly proportional to V_{ref}

B. inversely proportional to V_{ref}

C. independent of V_{ref}

D. directly proportional to clock frequency

Q6.11.1 MCQ 2011 1M

A 4-bit successive approximation type of A/D converter has an input range of 0 to 15 volts. The output bit b_1 next to the LSB has a stuck-at-zero fault. The pair of input voltages that produces the same output code word is

A. 2 V and 4 V **B.** 4 V and 6 V
C. 1 V and 2 V **D.** 8 V and 9 V

Q6.11.2 NAT 2011 1M

Figure below shows a circuit of implementing an 8-bit DAC using two identical 4-bit DACs with equal reference voltages. Assume that b_0 represents LSB, b_7 represents MSB and the opamp is ideal. To obtain correct analog values corresponding to an 8-bit DAC at the output V_o the value of resistor R is ____ k Ω .

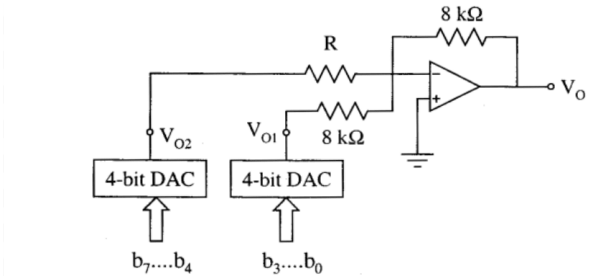
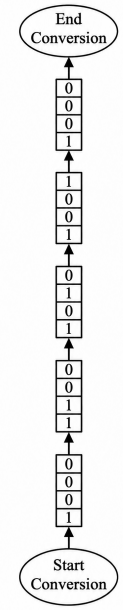


Fig. Q6.11.2

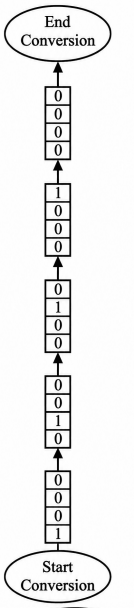
Q6.10.1 MCQ 2010 1M

A 4-bit successive approximation type ADC has full scale value of 15 V. The sequence of the states the SAR will traverse for the conversion of an input of 8.15 V is

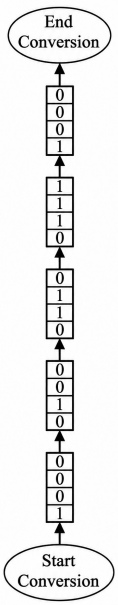
A.



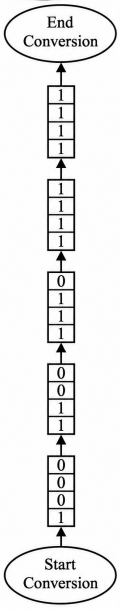
B.



C.



D.



Q6.09.1 MCQ 2009 1M

An 8-bit ADC, with 2's complement output has a nominal input range of -2 V to +2 V . It generates a digital code of 00 H for an analog input in the range -7.8125 mV to +7.815 mV. An input of -1.5 V will produce a digital output of

A. 90 H **B.** 96 H **C.** 9B H **D.** A0 H

Q6.08.1 MCQ 2008 1M

If the multiplexer is controlled such that the channels are sequenced every 5 μs as 1 , 2 , 1 , 3 , 1 , 4 , 1 , 2 , 1 , 3 , 1 , 4 , 1 , ..., the input connected to channel 1 will be sampled at the rate of

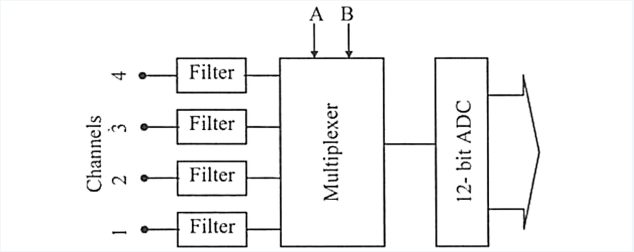
A. 25k samples/s **B.** 50k samples/s
C. 100k samples/s **D.** 200k samples/s

Common Data Questions

The next 2 questions share the data below.

Answer the next two questions using the following data.

A data acquisition sstem (DAS) shown below employs a successive approximation type 12-bit ADC having a conversion time of 5 μsec.



Q6.08.2 MCQ 2008 1M

The quantization error of the ADC is:

A. 0 % **B.** ± 0.012 % **C.** ± 0.024 % **D.** ± 0.048 %

Q6.08.3 MCQ 2008 1M

The system is used as a single channel DAS with channel 1 selected as input to the ADC which is set in the continuous conversion mode. For avoiding aliasing error, the cutoff frequency f_c of the filter in channel 1 should be

A. f_c < 100 kHz **B.** f_c = 100 kHz
C. 100 kHz < f_c < 200 kHz **D.** f_c = 200 kHz

Q6.07.1 MCQ 2007 1M

The circuit shown in the figure below works as a 2-bit ADC for $0 \le V_{in} \le 3$ V.

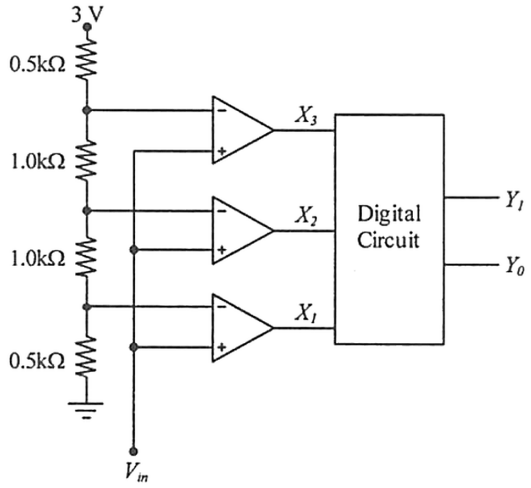


Fig. Q6.07.1

The MSB of the output Y_1 expressed as a Boolean function of the inputs X_1 , X_2 and X_3 is given by

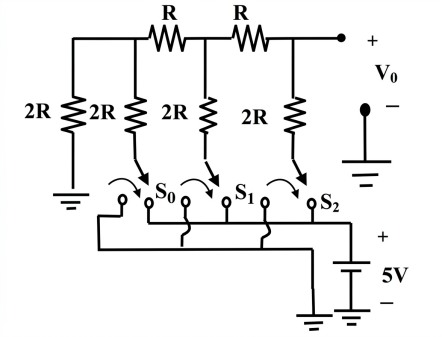
A. X_1 **B.** X_2 **C.** X_3 **D.** $X_1 + X_2$

Common Data Questions

The next 2 questions share the data below.

Answer the next two questions using the following data.

An R-2R ladder type DAC is shown. If a switch status is '0' , 0 V is applied and if a switch status is '1' , 5 V is applied to the corresponding terminal of the DAC.



Q6.06.1 MCQ 2006 1M

What is the output voltage (V_0) for the switch status $S_0 = 0$, $S_1 = 1$, $S_2 = 1$?

A. $\frac{5}{4}$ V **B.** $\frac{15}{4}$ V **C.** $\frac{17.4}{4}$ V **D.** $\frac{22.5}{4}$ V

Q6.06.2 MCQ 2006 1M

What is the step size of the DAC?

A. 0.125 V **B.** 0.525 V **C.** 0.625 V **D.** 0.75 V

Q6.06.3 MCQ 2006 1M

A single channel signal acquisition system with 0-10 V range consists of a sample-and-hold circuit with worst case drop rate of 100 μV/ms and 10-bit ADC. The maximum conversion time for the ADC is

A. 49 μs **B.** 0.49 ms **C.** 4.9 ms **D.** 49 ms

Q6.05.1 MCQ 2005 1M

For N-bit successive approximation ADCs, other parameters such as clock frequency remaining constant, the conversion time is proportional to

A. N^2 **B.** $\sqrt{N}$ **C.** log N **D.** N

Q6.04.1 MCQ 2004 1M

The minimum number of comparators required to build an 8-bit flash ADC is

A. 8 **B.** 64 **C.** 255 **D.** 256

Q6.03.1 MCQ 2003 1M

In a dual-slope ADC, the reference voltage is 100 mV and the first integration period is set as 50 ms. The input resistor of the integrator is 100 kΩ and the integrating capacitor is 0.04 μ F. For an input voltage of 120 mV, the second intergration (de-integration) period will be

A. 50 ms **B.** 60 ms **C.** 100 ms **D.** 120 ms

Q6.99.1 MCQ 1999 1M

In the figure the switches are in position a or b depending on the switch input being 0 or 1. B_3 is the MSB and B_0 is the LSB. When the input digital code is 1100 then output voltage V_0 is

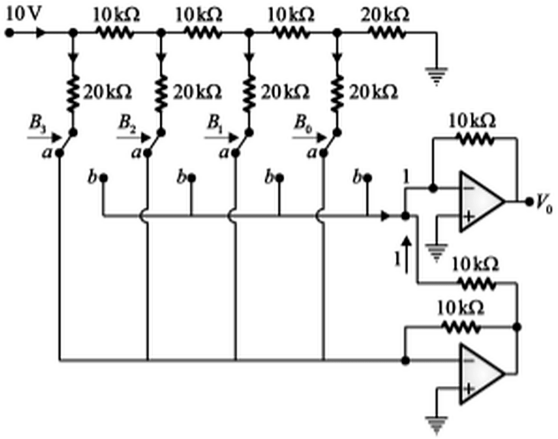


Fig. Q6.99.1

A. 5 V **B.** -5 V **C.** 6 V **D.** 7 V

Q6.99.2 MCQ 1999 1M

The conversion time of an 8-bit successive approximation ocnverter with a 1 MHz clock is nearly

A. 512 μs **B.** 256 μs **C.** 128 μs **D.** 8 μs

Q6.99.3 MCQ 1999 1M

The advantage of a dual slope converter over successive approximation converter is that the dual slope converter ____ .

A. is faster

B. eliminates error due to drift

C. can reduce errors due to power supply

D. does not require a stable voltage source

Q6.98.1 MCQ 1998 1M

Percent resolution of an 8 bit D/A converter is

A. 0.39 **B.** 0.78 **C.** 2.56 **D.** None of these

Q6.98.2 MCQ 1998 1M

The full scale input voltage to an ADC is 10 V. The resolution required is 5 mV. The minimum number of bits required for ADC is

A. 8 **B.** 10 **C.** 11 **D.** 12

Q6.98.3 MCQ 1998 1M

Integrated output waveform for the dual slope ADC is shown in the figure. The time T for an 8-bit counter with 4 MHz clock will be

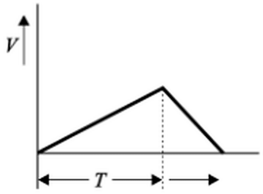


Fig. Q6.98.3

A. 0.032 ms **B.** 0.064 ms **C.** 0.64 ms **D.** 0.024 ms

Q6.97.1 MCQ 1997 1M

A sample-and-hold circuit is normally required before the following type of A/D converter

A. Successive approximation

B. Flash (parallel) converter

C. Voltage-to-frequency converter

D. Dual slope integrator

Q6.96.1 MCQ 1996 1M

A 3-bit resistance ladder D/A (R-2R network) has resistor values of R = 10 kΩ and $2R$ = 20 kΩ . V_{ref} equals 8 V. What is I_{out} for a digital input of 111?

A. 0.7 mA **B.** 0.2 mA **C.** 0.1 mA **D.** 1 mA

Q6.96.2 MCQ 1996 1M

An 8 bit ADC outputs all 1s when V_{in} = 1.275 V. The quantization error is

A. +5 mV **B.** -5 mV **C.** 10 mV **D.** ±2.5 mV

Q6.94.1 NAT 1994 1M

The percent resolution of a 10 bit D/A converter is ____ (round off to three decimal places).

Answer Key

Q6.26.1 63	Q6.25.1 A	Q6.25.2 125
Q6.24.1 3	Q6.22.1 A	Q6.22.2 10
Q6.21.1 4.9 to 5.1	Q6.21.2 A	Q6.21.3 B , D
Q6.20.1 3.5 to 4.5	Q6.20.2 223.3 to 223.5	Q6.19.1 64
Q6.18.1 C	Q6.14.1 A	Q6.14.2 A
Q6.11.1 C	Q6.11.2 0.5	Q6.10.1 A
Q6.09.1 D	Q6.08.1 C	Q6.08.2 B
Q6.08.3 D	Q6.07.1 B	Q6.06.1 B
Q6.06.2 D	Q6.06.3 D	Q6.05.1 D
Q6.04.1 C	Q6.03.1 B	Q6.99.1 B
Q6.99.2 D	Q6.99.3 C	Q6.98.1 A
Q6.98.2 C	Q6.98.3 B	Q6.97.1 A
Q6.96.1 A	Q6.96.2 D	Q6.94.1 0.098

PrepFusion

UNIT

VII

Logic Families and Semiconductor Memories

Topics

1. Logic Gate Implementation Using CMOS
2. Characteristics of Logic Families
3. Hazards in Combinational Circuits
4. Programmable Logic Devices (PLD, PLA, PAL)
5. ROM, SRAM and DRAM

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 10	**MCQ:** 9	**MSQ:** 1	**NAT:** 0
1M: 10	**2M:** 0	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2026)–1M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q7.26.1 MSQ 2026 1M

Which of the following functions represents the output of the CMOS logic gate shown in the circuit below?

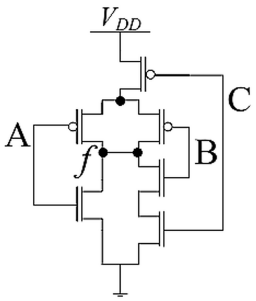


Fig. Q7.26.1

A. $f = \bar{A} + \bar{B}.\bar{C}$
B. $f = \overline{A + BC}$
C. $f = \bar{A}.(\bar{B} + \bar{C})$
D. $f = \overline{A.(B + C)}$

Q7.23.1 MCQ 2023 1M

Choose the fastest logic family among the following:

A. Transistor-Transistor Logic

B. Emitter-Coupled Logic

C. CMOS Logic

D. Resistor-Transistor Logic

Q7.15.1 MCQ 2015 1M

For the circuit shown in the figure, the raising edge triggered D-flipflop with asynchronous reset has a clock frequency of 1 Hz. The NMOS transistor has an ON resistance of 1000 Ω and an OFF resistance of infinity.

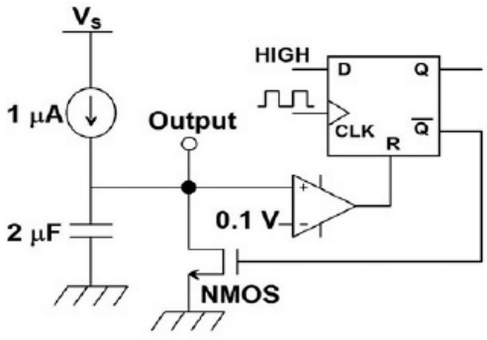


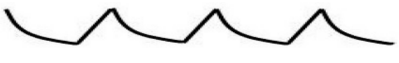
Fig. Q7.15.1

The nature of the output waveform is:

A.



B.



C.



D.



Q7.14.1 MCQ 2014 1M

The figure is a logic circuit with inputs A and B and output Y. V_{SS} = +5 V. The circuit is of type

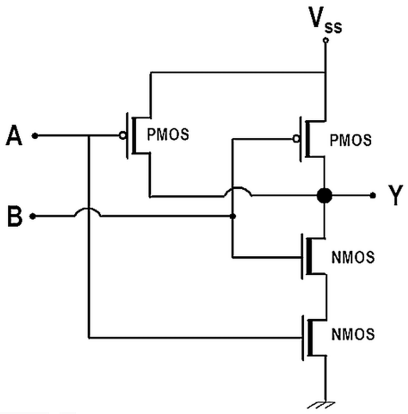


Fig. Q7.14.1

A. NOR **B.** AND **C.** OR **D.** NAND

Q7.07.1 MCQ 2007 1M

A CMOS implementation of a logic gate is shown in the following figure:

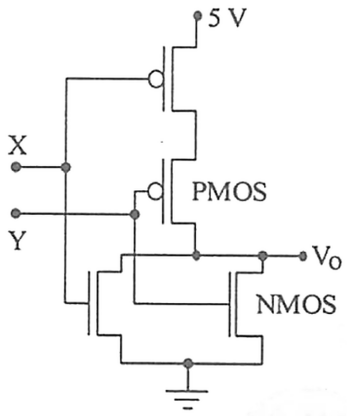


Fig. Q7.07.1

The Boolean logic realised by the circuit is

A. AND **B.** NAND **C.** NOR **D.** OR

Q7.04.1 MCQ 2004 1M

The sink current of the TTL NAND gate shown below is 16 mA and the input capacitance of the CMOS NAND gate is 10 pF. The minimum pull-up time constant in ns for the CMOS gate with these specifications is

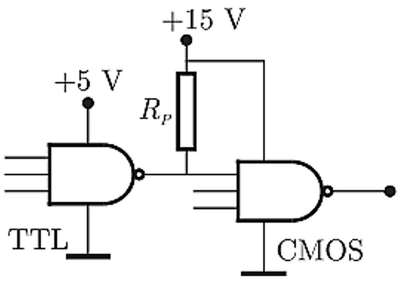


Fig. Q7.04.1

The pull-up supply is +12 V.

A. 10.2 **B.** 7.5 **C.** 8.5 **D.** 6.5

Q7.04.2 MCQ 2004 1M

In figure, if the input C = 0 and D = 0 , the output Y can be made 0 by making

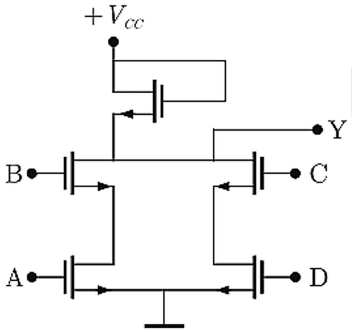


Fig. Q7.04.2

A. A = 0 , B = 1 **B.** A = 0 , B = 0
C. A = 1 , B = 1 **D.** A = 1 , B = 0

Q7.03.1 MCQ 2003 1M

For the CMOS analog switch shown in figure, the positive supply is $+V_{DD}$ and the negative supply is $+V_{EE}$. The input V_i is bipolar. The switch will be ON and V_0 will be equal to V_i provided

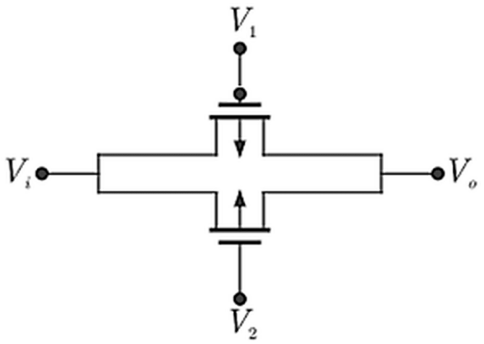


Fig. Q7.03.1

A. $V_1 = V_2 = +V_{DD}$

B. $V_1 = V_2 = -V_{EE}$

C. $V_1 = +V_{DD}$ and $V_2 = -V_{EE}$

D. $V_1 = -V_{EE}$ and $V_2 = +V_{DD}$

Q7.03.2 MCQ 2003 1M

Introducing a schottky diode between the base and collector of the output transistor in a TTL circuit:

A. Increases the speed of operation by inhibiting saturation

B. Decreases the speed of operation by inhibiting saturation

C. Increases the FAN OUT by enabling saturation

D. Increases the speed of operation by enabling saturation

Q7.94.1 MCQ 1994 1M

The diode circuit shown in figure functions as

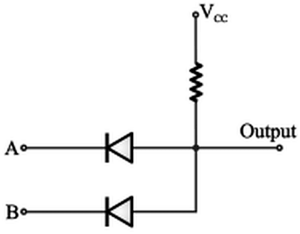


Fig. Q7.94.1

A. AND gate **B.** OR gate
C. NAND gate **D.** NOR gate

Answer Key

Q7.26.1 B , C	Q7.23.1 B	Q7.15.1 A
Q7.14.1 D	Q7.07.1 C	Q7.04.1 B
Q7.04.2 C	Q7.03.1 D	Q7.03.2 A
Q7.94.1 A		

PrepFusion

UNIT

VIII

Computer Organization and Architecture

Topics

1. Machine Instructions and Addressing Modes
2. ALU, Data Path and Control Unit
3. Instruction Pipelining

PrepFusion

Unit Snapshot*

Stream: IN	**Questions:** 48	**MCQ:** 41	**MSQ:** 3	**NAT:** 3
1M: 45	**2M:** 3	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2018)–3M(2010)

**See preface for how Avg Weightage and Range are calculated.*

Q8.26.1 MSQ 2026 2M

A 16-bit processor has 16-bit wide instructions and a few internal registers. Every 16-bit instruction is encoded in a fixed format with one operation code of 7 bits and up to three operands. The operand can be an internal register and each operand is encoded by 3 bits. All the internal registers can be specified in an instruction using 3 bits. The correct statements for this processor is/are :

A. The processor has 16 internal registers.

B. The processor can support 128 unique operation codes (Opcodes).

C. The processor has up to 8 internal registers.

D. The processor can support 512 unique operation codes (Opcodes).

Q8.21.1 MSQ 2021 2M

A 16-bit microprocessor has twenty address lines (A_0 to A_{19}) and 16 data lines. The higher eight significant lines of the data bus of the processor are tied to the 8 data lines of a 16 Kbyte memory that can store 1-byte in each of its 16 K address location. The memory chip should map onto contiguous memory locations and occupy only 16 Kbyte of memory space. Which of the following statement(s) is/are correct with respect to the above design?

A. If the 16 Kbyte of memory chip is mapped with a starting address of 80000H, then the ending address will be 83FFFH.

B. The active high chip-select needed to map 16 Kbyte memory with starting address at F0000H is given by the logic expression (A_{19} , A_{18} , A_{17} , A_{16}).

C. The 16 Kbyte memory cannot be mapped with continuous address location with a starting address as 0F000H using only A_{19} to A_{14} for generating chip select.

D. The above chip cannot be interfaced as the width of the data bus of the processor and memory chip differs.

Q8.19.1 NAT 2019 2M

In a microprocessor with a 16-bit address bus, the most significant address lines A15 to A12 are used to select a 4096 word memory unit, while lines A0 to A11 are used to address a particular word in the memory unit. if the 3 least significant lines of the address bus A0 to A2 are short-circuited to ground, the addressable number of words in the memory unit is ____ .

Q8.18.1 NAT 2018 1M

A portion of an assembly language porgram written for an 8-bit microprocessor is given below along with explanations. The code is intended to introduce a software time delay. The processor is driven by a 5 MHz clock. The time delay (in μs) introduced by the program is ____.

- MVIB, 64 H : Move immediate the given byte into register B. Takes 7 clock periods.
- LOOP : DCR B : Decrement register B. Affects Flags. Takes 4 clock periods.
- JNZ LOOP: Jump to address with label LOOP if zero flag is not set. Takes 10 clock periods when jump is performed and 7 clock periods when jump is not performed.

Q8.17.1 MCQ 2017 1M

An 8-bit microcontroller with 16 address lines has 3 fixed interrupts, i.e. , Int1, Int2 and Int3 with corresponding interrupt vector addresses as 0008H, 0010H and 0018H. To execute a 32-byte long Interrupt Service Subroutine for Int1 starting at the address IIS1, the location 0008H onwards should ideally contain

A. A CALL to ISS 1

B. An unconditional JUMP to ISS1

C. A conditional JUMP to ISS1

D. Only ISS1

Q8.16.1 MSQ 2016 1M

A 1 Kbyte memory module has to be interfaced with an 8-bit microprocessor that has 16 address lines. The address lines A_0 to A_9 of the processor are connected to the corresponding address lines of the memory. The active low chip select $\overline{CS}$ of the memory module is connected to the y_5 output of a 3-to-8 decoder with active low outputs. S_0 , S_1 and S_2 are the input lines to the decoder, with S_2 as the MSB. The decoder has one active low $\overline{EN_1}$ and one active high EN_2 enable lines as shown below.

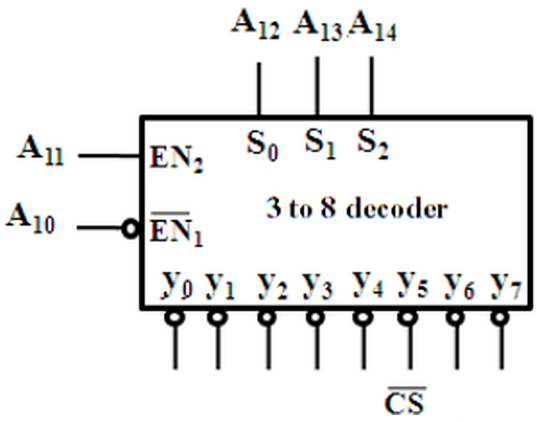


Fig. Q8.16.1

The address range(s) that gets mapped onto this memory module is/are

A. 3000H to 33FFH and E000h to E3FFH

B. 1400H to 17FFH

C. 5300H to 53FFH and A300h to A3FFH

D. 5800H to 5BFFH and D800H to DBFFH

Q8.15.1 MCQ 2015 1M

An ADC is interfaced with a microprocessor as shown in the figure. All signals have been indicated with typical notations.

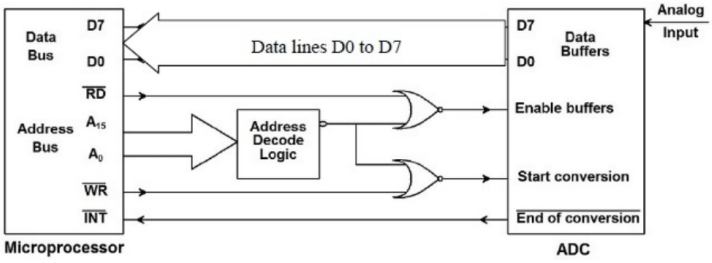


Fig. Q8.15.1

Acquisition of one new sample of the analog input signal by the microprocessor involves

A. One READ cycle only

B. One WRITE cycle only

C. One WRITE cycle followed by one READ cycle

D. One READ cycle followed by one WRITE cycle

Q8.14.1 MCQ 2014 1M

A microprocessor accepts external interrupts (Ext INT) through a programmable interrupt controller as shown in the figure.

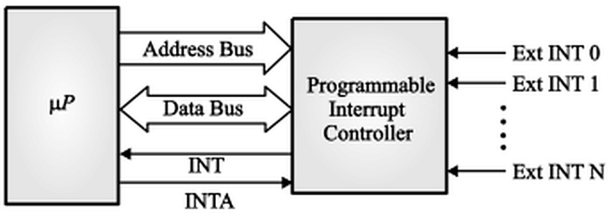


Fig. Q8.14.1

Assuming vectored interrupt, a correct sequence of operations when a single external interrupt (Ext INT) is received will be:

A. Ext INT1 → INTA → Data Read → INT

B. Ext INT1 → INT → INTA → Data Read

C. Ext INT1 → INT → INTA → Address Write

D. Ext INT1 → INT → Data Read → Address Write

Q8.11.1 MCQ 2011 1M

An 8K × 8 bit RAM is interfaced to an 8085 microprocessor. In a fully decoded scheme if the address of the last memory location of this RAM is 4FFFH, the address of the first memory location of the RAM will be

A. 1000H **B.** 2000H **C.** 3000H **D.** 4000H

Q8.11.2 MCQ 2011 1M

The number of objects crossing a window sequentially at variable speed is to be counted using an interrupt in the 8085 microprocessor. The objects are sensed by an optical source and a detector with associated signal conditioning circuit. The circuit produces a logic high output as long as the object is in front of the window, and this output is used to interrupt the processor. The duration of an object being in front of the window is in the range of 100 ms to 2s. The processor takes 1 ms to process the input after an interrupt. The best choice of interrupt for an error fee counting is

A. RST 5.5 **B.** RST 6.5 **C.** RST 7.5 **D.** INTR

Q8.10.1 MCQ 2010 1M

An 8-bit DAC is interfaced with a microprocessor having 16 address lines (A_0 ... A_{15}) as shown in the adjoining figure. A possible valid address for this DAC is

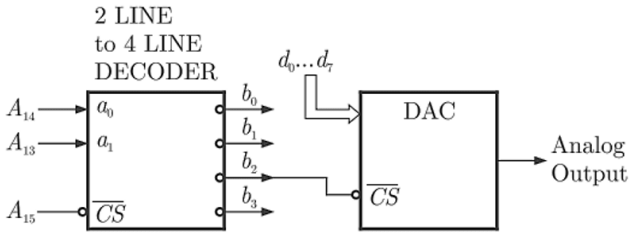


Fig. Q8.10.1

A. 3000H **B.** 4FFFH **C.** AFFFH **D.** C000H

Q8.10.2 MCQ 2010 1M

In a 8085 processor, the main program calls the subroutine SUB1 given below. When the program returns to the main program after executing SUB1, the value in the accumulator is

Address	Opcode	Mnemonic
2000	3E00	SUB1: MVI A, 00H
2002	CD 05 20	CALL SUB2
2005	3C	SUB2: INR A
2006	C9	RET

Accumulator for A = 00 increases and finally

A. 00 **B.** 01 **C.** 02 **D.** 03

Q8.10.3 MCQ 2010 1M

The subroutine SBX given below is executed by an 8085 processor. The value in the accumulator immediately after the execution of the subroutine will be

```
SBX:  MVI A, 99H
      ADI 11H
      MOV C, A
      RET
```

A. 00H **B.** 11H **C.** 99H **D.** AAH

Q8.09.1 MCQ 2009 1M

Consider a system consisting of a microprocessor, memory and peripheral devices connected by a common bus. During DMA data transfer, the microprocessor

A. only reads from the bus

B. only writes to the bus

C. both reads from and writes to the bus

D. neither reads from nor writes to the bus

Q8.09.2 MCQ 2009 1M

The following is an assembly language program for 8085 microprocessors.

Address	Instruction code	Mnemonic
1000H	3E 06	MVI A, 06H
1002H	C6 70	ADI 70H
1004H	32 07 10	STA 1007H
1007H	AF	XRA A
1008H	76	HLT

When this program halts, the accumulator contains

A. 00H **B.** 06H **C.** 70H **D.** 76H

Q8.08.1 MCQ 2008 1M

A 2k x 8 bit RAM is interfaced to an 8-bit microprocessor. If the address of the first memory location in the RAM is 0800H, the address of the last memory location will be

A. 1000H **B.** 0FFFH **C.** 4800H **D.** 47FFH

Q8.07.1 MCQ 2007 1M

A snapshot of the address, data and control buses of an 8085 microprocessor executing a program is given below.

Address	2020H
Data	24 H
IO/$\overline{M}$	Logic High
$\overline{RD}$	Logic High
$\overline{WR}$	Logic Low

The assembly language instruction being executed is

A. IN 24 H **B.** IN 20 H
C. OUT 24 H **D.** OUT 20 H

Q8.07.2 MCQ 2007 1M

8-bit signed integers in 2's complement form are read into the accumulator of an 8085 microprocessor from an input/output port using the following assembly language program segment with symbolic addresses.

Label	Mnemonics
BEGIN :	IN PORT
	RAL
	JNC BEGIN
	RET
	RAR
END :	HLT

This program:

A. halts upon reading a negative number

B. halts upon reading a positive number

C. halts upon reading a zero

D. never halts

Q8.06.1 MCQ 2006 1M

A memory mapped input/output device has an address of 00FoH. Which of the following 8085 instructions outputs the content of the accumulator to the input/output device?

A. LXI H, 00FoH
MOV M, A

B. LXI H, 00FoH
OUT M

C. LXI H, 00FoH
OUT F0H

D. LXI H, 00FoH
MOV A, M

Q8.06.2 MCQ 2006 1M

An 8085 assembly language program is given as follows. The execution time of each instruction is given against the instruction in terms of T-state.

	Instruction	T-state
.....	.....	.....
	MVI B, 0A H	7T
LOOP :	MVI C, 05H	7T
	DCR C	4T
	DCR B	4T
	JNZ LOOP	10T/7T

The execution time of the program in terms of T-states is

A. 247 T **B.** 250 T

C. 254 T **D.** 257 T

Q8.05.1 MCQ 2005 1M

A microprocessor has an instruction XOR (r_1, r_2) , which performs an ExclusiveOR operation of registers r_1, r_2 and stores the result in r_1. After the following instructions are executed:

XOR (r_2, r_1)
XOR (r_1, r_2)
XOR (r_2, r_1)

Which one of the following is true?

A. Content of register r_1 is half sum of r_1 and r_2

B. Content of register r_2 is half sum of r_1 and r_2

C. Contents of registers r_1 and r_2 remain unaltered

D. Contents of registers r_1 and r_2 are swapped

Q8.05.2 MCQ 2005 1M

The time period of a square wave in the audio frequency range is measured using an 8085 microprocessor by feeding the square wave to one of the four interrupts, namely RST 7.5, RST 6.5, RST 5.5, or INT. The algorithm used starts a timer at the beginning of a time period, stops the timer at the beginning of the next time period and reads the timer values for time measurement. Which of the following interrupts should be selected for this application?

A. INTR **B.** RST 5.5 **C.** RST 6.5 **D.** RST 7.5

Q8.05.3 MCQ 2005 1M

An 8-bit microcontroller has an external RAM in the memory map from 8000H to 9FFFH. The number of bytes this RAM can store, is

A. 8193 **B.** 8191 **C.** 8192 **D.** 8000

Q8.05.4 MCQ 2005 1M

In an 8085 microprocessor, which one of the following is the correct sequence of the machine cycles for the execution of the DCR M instruction?

A. Op-code fetch

B. Op-code fetch, memory read, memory write

C. Op-code fetch, memory read

D. Op-code fetch, memory write, memory write

Q8.05.5 MCQ 2005 1M

In a 8085 microprocessor, the value of the Stack Pointer (SP) is 2010 H and that of DE register pair is 1234 H before the following code is executed. The value of DE register pair after the following code is executed

```
LXI H, 0000H
PUSH H
PUSH H
POP B
DADA SP
XCHG
```

A. 200E H **B.** 200C H **C.** 2010 H **D.** 1232 H

Q8.04.1 MCQ 2004 1M

The vectored address corresponding to the software interrupt command RST7 in 8085 microprocessor is

A. 0017 H **B.** 0027 H **C.** 0038 H **D.** 0700 H

Q8.04.2 MCQ 2004 1M

The following 8085 instructions are executed sequentially

```
PROG  XRA  A
      MOV  L,A
      MOV  H,L
      INX  H
      DAD  H
```

After execution, the content of HL register pair is

A. 0000 H **B.** 0101 H **C.** 0001 H **D.** 0002 H

Q8.03.1 Descriptive 2003 1M

Some of the pins of an 8085 CPU and their functions are listed below. Identify the correct answer that matches the pins to their respective functions.

P. RST 7.5	1. Selects I/O or memory
Q. HOLD	2. Demultiplexes the address and data bus
R. IO/M	3. Is a vectored interruptL
S. ALE	4. Facilitates direct memory access
	5. Is a clock
	6. Selects BCD mode of operation

Codes:	P	Q	R	S
A.	3	2	1	4
B.	4	1	5	3
C.	3	4	1	2
D.	2	3	6	1

Q8.03.2 MCQ 2003 1M

In an 8085 system containing 8 kB of ROM and k8 byte of RAM, the ROM is selected when A_{15} is 0 and the RAM is selected when A_{15} is 1. A_{13} and A_{14} are unused. The CPU executes the following program:

```
Prog1  MVI A, 00H
       STA 8080H
       DCR A
       STA C080H
       RET
```

The content of memory location 8080H after the execution of the RETURN instruction is

A. FFH **B.** FEH **C.** 01H **D.** 00H

Q8.03.3 MCQ 2003 1M

A ROM is interfaced to an 8085 CPU as indicated in figure. The address range occupied by the ROM is

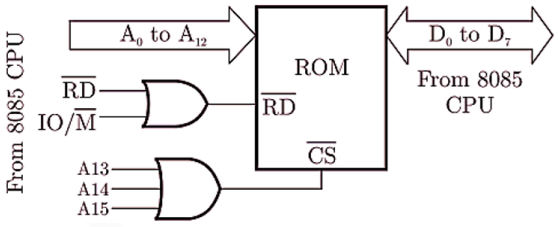


Fig. Q8.03.3

A. 0000 - 0FFF **B.** 0000 - 1FFF
C. 0000 - 2FFF **D.** 8000 - 9FFF

Q8.02.1 MCQ 2002 1M

In an INTEL 8085 microprocessor the ADDRESS-DATA bus and the DATA bus are

A. non-multiplexed **B.** multiplexed
C. duplicated **D.** same as CONTROL bus

Q8.02.2 MCQ 2002 1M

The 14-bit timer of 8156 is loaded with the counter value of 07DOH. The timer input is connected to a programmed to produce a continuous signature wave output. The frequency of the square wave output is

A. 400 kHz **B.** 800 kHz **C.** 400 Hz **D.** 2000 kHz

Q8.02.3 MCQ 2002 1M

In an 8085 based system, the subroutine TEST given below is called by another program. When the processor returns from the subroutine TEST, the value in the accumulator will be

```
TEST   MOV A, 00H
       CALL TESK
TESK   INR A
       RET
```

A. 02 **B.** 00 **C.** FF **D.** 20

Q8.01.1 MCQ 2001 1M

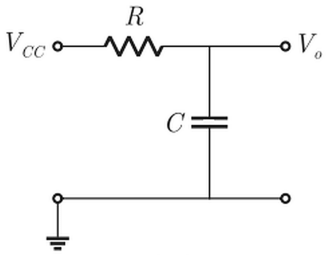
An m-bit microprocessor has an m-bit

A. flag register **B.** instruction register
C. data counter **D.** program counter

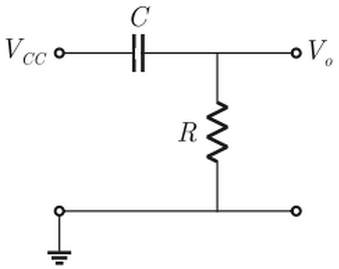
Q8.01.2 MCQ 2001 1M

In figure, identify the circuit which is appropriate for use as a power-on reset for a microprocessor.

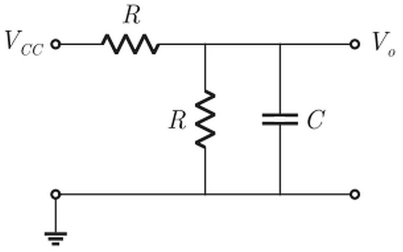
A.



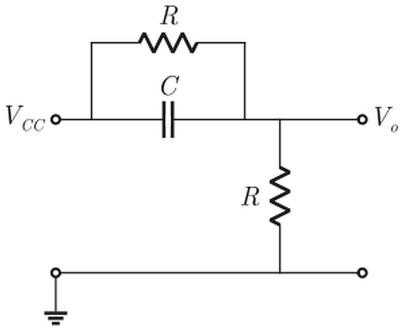
B.



C.



D.



Q8.01.3 MCQ 2001 1M

Microprocessor 8085 regains control of the bus

A. immediately after HOLD goes low

B. immediately after HOLD goes high

C. after half-clock cycle after HLDA goes low

D. after half-clock cycle after HLDA goes high

Q8.01.4 MCQ 2001 1M

In a microprocessor with 16 address and 12 data lines, the maximum number of opcodes is

A. 2^6 **B.** 2^8 **C.** 2^{12} **D.** 2^{16}

Q8.01.5 MCQ 2001 1M

In 8085 microprocessor, CY flag may be set by the instruction

A. SUB **B.** INX **C.** CMA **D.** ANA

Q8.00.1 MCQ 2000 1M

For an 8085 microprocessor, the Stack Pointer (SP) and Program Counter (PC) registers contain the number F000 and 2400 in Hex respectively. The contents of the register after execution of the instruction CALL E000 would be

A. PC : F003 , SP : 2400

B. PC : E000 , SP : 2400

C. PC : E000 , SP : 2401

D. PC : E000 , SP : 23 FE

Q8.00.2 MCQ 2000 1M

Indicate which one of the following statements is correct.

A. Static RAMs are faster than dynamic RAMs

B. RAMs cannot be used to realise Read Write Memory (RWM)

C. ROMs are not random access devices

D. RAMs are generally non-volatile

Q8.00.3 MCQ 2000 1M

Find the correct match among the following pairs in the context of an 8085 microprocessor.

(a)	DAA	(e)	Program control instruction
(b)	LXI	(f)	Data movement instruction
(c)	RST	(g)	Interrupt instruction
(d)	JMP	(h)	Arithmetic

A. a-e, b-f, c-g, d-h

B. a-h, b-f, c-g, d-e

C. a-h, b-g, c-f, d-e

D. a-f, b-h, c-g, d-e

Q8.99.1 MCQ 1999 1M

The important feature of microcontroller is that it has on-chip

A. math co-processor **B.** program memory
C. interface for I/O devices **D.** hardware multiplier

Q8.99.2 MCQ 1999 1M

A computer has a memory space of 2^{16} and word length of 24 bits. The memory chips available have 10 address and 8 data lines. The number of chips required for the computer memory space is

A. 192 **B.** 256 **C.** 512 **D.** 1024

Q8.98.1 MCQ 1998 1M

Identify the software interrupt instruction in 8085 microprocessor

A. INT **B.** RST 5 **C.** RST 7.5 **D.** RST 6.5

Q8.97.1 MCQ 1997 1M

The stack pointer in a microprocessor is a register containing

A. the address of the next operand

B. the current size of the stack

C. the address of the top of the stack

D. the address for storing the result of arithmetic operations

Q8.94.1 MCQ 1994 1M

In a 8-bit microprocessor with 16 address lines, the range of signed integers that can be processed is

A. -127 to 127 **B.** -125 to 125
C. -255 to 255 **D.** -256 to 256

Q8.94.2 NAT 1994 1M

The program counter on a 8-bit microprocessor is always a ____-bit register.

Q8.94.3 MCQ 1994 1M

The contents of the STACK memory in a microprocessor system are retrieved on

A. first in first out basis

B. last in last out basis

C. last in first out basis

D. both (A) and (B)

Answer Key

Q8.26.1 B , C	Q8.21.1 A, C	Q8.19.1 512
Q8.18.1 279 to 282	Q8.17.1 B	Q8.16.1 D
Q8.15.1 C	Q8.14.1 C	Q8.11.1 C
Q8.11.2 C	Q8.10.1 A	Q8.10.2 C
Q8.10.3 D	Q8.09.1 D	Q8.09.2 D
Q8.08.1 B	Q8.07.1 D	Q8.07.2 A
Q8.06.1 A	Q8.06.2 C	Q8.05.1 D
Q8.05.2 D	Q8.05.3 C	Q8.05.4 B
Q8.05.5 A	Q8.04.1 C	Q8.04.2 D
Q8.03.1 C	Q8.03.2 D	Q8.03.3 B
Q8.02.1 B	Q8.02.2 C	Q8.02.3 A
Q8.01.1 B	Q8.01.2 B	Q8.01.3 C

Q8.01.4 C	Q8.01.5 A	Q8.00.1 D
Q8.00.2 A	Q8.00.3 B	Q8.99.1 C
Q8.99.2 A	Q8.98.1 B	Q8.97.1 C
Q8.94.1 A	Q8.94.2 16	Q8.94.3 C

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